

Construction of Higher Sheaf Theory

by

Yang Hu

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by Yang Hu

Department of Mathematics

The Hong Kong University of Science and Technology

Abstract

In this thesis, we discuss a construction of higher sheaf theory. Usual sheaf theory can be encoded as a lax symmetric monoidal $(\infty, 1)$ -functor from the category of spans in the category of schemes or stacks to the ∞ -category of presentable categories:

$$\mathrm{Shv}(-) : \mathrm{Span}_1(\mathcal{C}) \rightarrow \mathrm{Pr}^{\mathrm{L}}.$$

Here $\mathcal{C}$ denotes the category of schemes or stacks. Starting from a given sheaf theory $\mathrm{Shv}(-)$, we construct an $(\infty, 2)$ lax symmetric monoidal functor

$${}^2\mathrm{Shv}(-) : \mathrm{Span}_2(\mathcal{C}) \rightarrow \mathrm{LMod}_{1\mathrm{Pr}}(1\mathrm{Pr}) = {}^2\mathrm{Pr},$$

which may be regarded as a higher sheaf theory. And as an example, we discuss higher sheaf on the classifying stack BG of a smooth algebraic group scheme G .

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This is to certify that I have examined the above Master of Philosophy thesis
and have found that it is complete and satisfactory in all respects,
and that any and all revisions required by
the thesis examination committee have been made.

Quoc Ho, Thesis Supervisor

Xu Kun, Head of Department

Department of Mathematics
5 July 2026

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1 Introduction

1.1 Motivation

Braided monoidal category structure appears as a fundamental framework to understand knot theory and topological field theory; it can be viewed as a special case of $\mathbb{E}_2$ algebra object in the $(2, 1)$ category $\text{Cat}_{\text{ordinary}}$ of ordinary categories. By Dunn additivity, we can characterize the $(\infty, 1)$ operad $\mathbb{E}_2$ as the Boardman-Vogt tensor product $\mathbb{E}_1 \otimes \mathbb{E}_1$ of $\mathbb{E}_1$ operad.

There is an ongoing project on $(\infty, 2)$ operad theory, developed by Tallak Manum, and by taking the Gray-Boardman-Vogt tensor product of $\mathbb{E}_1$ with itself, we get an $(\infty, 2)$ operad $\mathbb{E}_2^{\text{lax}}$. Moreover, one can characterize the $\mathbb{E}_2^{\text{lax}}$ algebra in the symmetric monoidal $(\infty, 2)$ category $\text{Span}_2(\mathcal{C})$ in a nice way, by the work of Quoc Ho, Walker H. Stern, and Jonte Gödicke. The space of $\mathbb{E}_2^{\text{lax}}$ objects in $\text{Span}_2(\mathcal{C})$ is precisely the space of 2-Segal objects in $\mathcal{C}$, which is a ‘lax’ and ‘ $\mathbb{E}_2$ ’ generalization of the results of the $\mathbb{E}_1$ algebra object in $\text{Span}_1(\mathcal{C})$ in the PHD thesis of Jonte Gödicke.

To transfer the $\mathbb{E}_2^{\text{lax}}$ algebra objects, we need the ‘lax symmetric monoidal $(\infty, 2)$ functor’, which is a generalization of the lax symmetric monoidal $(\infty, 1)$ functor developed in [25] by Lurie. The symmetric monoidal $(\infty, 2)$ functor can be defined as the morphism in the category $\text{CAlg}(\text{Cat}_2)$ where Cat_2 is equipped with the Cartesian symmetric monoidal structure. By the (Un)straightening of $(\infty, 2)$ categories, any symmetric monoidal $(\infty, 2)$ category $\mathcal{C}$ unstraightens to an $(\infty, 2)$ coCartesian fibration $\mathcal{C}^\otimes$ over $\text{Comm}^\otimes$, and the symmetric monoidal functor is a morphism between the associated $(\infty, 2)$ coCartesian fibrations.

Definition 1.1.1. A 2 functor $\pi : C \rightarrow D$ is a 2-coCartesian fibration if it satisfies the condition:

- (coCartesian 1 morphism) Given any $c_1 \in C$, and 1 morphism $e : \pi(c_1) \rightarrow d$ in D , there exists a 1 morphism $f : c_1 \rightarrow c_2$ such that $\pi(f) = e$ and f induces a pullback

$$\begin{array}{ccc} \text{Map}_C(c_2, x) & \longrightarrow & \text{Map}_C(c_1, x) \\ \downarrow & \lrcorner & \downarrow \\ \text{Map}_D(\pi(c_2), \pi(x)) & \longrightarrow & \text{Map}_D(\pi(c_1), \pi(x)) \end{array} \quad \text{for any } x \in C.$$

diagram:

- (Cartesian 2 morphism) $\text{Map}_C(x, y) \rightarrow \text{Map}_D(\pi(x), \pi(y))$ is a Cartesian fibration.
- The composition map $\text{Map}_C(x, y) \times \text{Map}_C(y, z) \rightarrow \text{Map}_C(x, z)$ preserves Cartesian edges.

And we can define the lax symmetric monoidal $(\infty, 2)$ functor as

Definition 1.1.2. A lax symmetric monoidal $(\infty, 2)$ functor $G : \mathcal{C}^\otimes \rightarrow \mathcal{D}^\otimes$ is a functor that preserves the inert coCartesian 1 morphisms lifting and Cartesian 2 morphisms.

By [2], we have:

Theorem 1.1.3. Given an adjunction $f : \mathcal{C} \rightleftharpoons \mathcal{D} : g$ in Cat_2 , if the left adjoint f can be enhanced into a symmetric monoidal $F : \mathcal{C}^\otimes \rightarrow \mathcal{D}^\otimes$ in $\text{CAlg}(\text{Cat}_2)$, then the right adjoint can be enhanced into a lax symmetric monoidal $(\infty, 2)$ functor $G : \mathcal{D}^\otimes \rightarrow \mathcal{C}^\otimes$.

On the other hand, a sheaf theory can be encoded into a lax symmetric monoidal functor $\text{Shv}(-) :$

$\text{Span}_1(\mathcal{C}) \rightarrow \text{Pr}^L$, in the recent work on higher sheaf theory by Peter Scholze and H erman Stefanich, for stacky objects, the higher sheaf theory recovers more geometric information than 1 sheaf theory. In this thesis, we construct a lax symmetric monoidal $(\infty, 2)$ functor $2\text{Shv}(-) : \text{Span}_2(\mathcal{C}) \rightarrow 2\text{Pr}_\kappa$ for some fixed regular cardinality κ , assigning the higher sheaf. And we can construct interesting $\mathbb{E}_2$ and $\mathbb{E}_2^{lax}$ algebra objects in 2Pr_κ through the functor $2\text{Shv}(-)$.

1.2 A sketch of the thesis

In section 2, we supplement the details of [37] by Stefanich, showing that $\text{Span}_n(\mathcal{C}) \in \text{Cat}_n$ can be enhanced into an object in $n\text{Cat}^{\text{Span}_1(\mathcal{C})}$, denoted as $\text{Span}_n^{\text{enr}}(\mathcal{C})$. In section 3, we show that there exists a lax symmetric monoidal (∞, n) functor from $\text{Span}_n^{\text{enr}}(\mathcal{C}) \rightarrow (n-1)\text{Cat}^{\text{Span}_1(\mathcal{C})}$. As an application, in section 4, we develop an $(\infty, 2)$ six functor formalism $2\text{Shv}(-) : \text{Span}_2(\mathcal{C}) \rightarrow 2\text{Pr}$ from a given six functor formalism $\text{Shv}(-) : \text{Span}_1(\mathcal{C}) \rightarrow 1\text{Pr}$, and discuss its value on the classifying stack BG of a smooth algebraic group scheme. And in the appendix, there are some necessary background materials on enriched ∞ -category theory.

1.3 Notation and convention

1.3.1 Category

- Let Cat_n denote the ∞ -category of (∞, n) categories. When $n = 1$, let Cat denote Cat_1 .
- Let Cat^{pb} denote the ∞ -categories with fiber products and ∞ -functors preserving fiber products.
- Let Spc denote the ∞ -category of Kan complexes.
- Given an ∞ -category $\mathcal{C}$, let $\mathcal{C}^\simeq$ denote the maximal Kan complex of $\mathcal{C}$.
- Let Pr^L denote the ∞ -category of presentably ∞ -categories as defined in [24, definition 4.4.0.1].
- Let 1Pr denote the ∞ -category of $\aleph_1$ -compactly generated categories as defined in [24, definition 5.5.7.1], which itself is a presentably category.

1.3.2 Inclusions and embeddings

- We often call a fully faithful functor $\mathcal{C} \rightarrow \mathcal{D}$ an embedding.
- We call a functor $\phi : \mathcal{D} \rightarrow \mathcal{D}$ an inclusion or say that ϕ exhibits $\mathcal{C}$ as a subcategory of $\mathcal{D}$ if for any ∞ -category E the induced map $\text{Cat}_\infty(E, \mathcal{C}) \rightarrow \text{Cat}_\infty(E, \mathcal{D})$ is an embedding. A functor $\phi : \mathcal{C} \rightarrow \mathcal{D}$ is an inclusion if and only if it induces an embedding on maximal subspaces and on all mapping spaces.
- We say that an inclusion $\phi : \mathcal{C} \rightarrow \mathcal{D}$ exhibits $\mathcal{C}$ as a wide subcategory of $\mathcal{D}$ if ϕ is essentially surjective.

1.3.3 Higher algebra

- Let Op be the category of operads as defined in [25, chapter 2].
- Let $\text{Ass} \cong \Delta^{op}$ be the approximation of the operad $\text{Assoc}^{\otimes}$ in [25, definition 4.1.1.3].
- Let Op^{ns} be the category of non-symmetric operads, as defined in [15, notation 2.36], which is equivalent to the slice category of Op over $\text{Assoc}^{\otimes}$ by [25, theorem 4.3.14].
- Let Mon^{lax} be the full category of Op^{ns} consisting of monoidal categories.
- Let Mon be the wide subcategory of Mon^{lax} with 1-morphisms being monoidal functors.
- Given two operads $\mathcal{V}$ and $\mathcal{W}$, let $\text{Alg}_{\mathcal{V}}(\mathcal{W})$ denote the ∞ -category of operad maps between $\mathcal{V}$ and $\mathcal{W}$. When $\mathcal{V} = \text{Comm}^{\otimes}$, let $\text{CAlg}(\mathcal{W})$ denote $\text{Alg}_{\mathcal{V}}(\mathcal{W})$.
- We call an object $\mathcal{C} \in \text{CAlg}(\text{Cat}_n)$ a symmetric monoidal (∞, n) category.
- We call a morphism $F : \mathcal{C} \rightarrow \mathcal{D}$ in $\text{CAlg}(\text{Cat}_n)$ a symmetric monoidal (∞, n) functor. Especially, the right adjoint of a symmetric monoidal (∞, n) functor is a lax symmetric monoidal (∞, n) functor.

2 $\text{Span}_n^{enr}(\mathcal{C}) \in n \text{Cat}^{\text{Span}_1(\mathcal{C})}$

2.1 Span_n

In this subsection, we review the constructions of $\text{Span}_n(\mathcal{C})$ and $\text{Span}_{n-1}^+(\mathcal{C})$ in [12, section 5].

Construction 2.1.1. Consider the following functor

$$\overline{\text{SPAN}}_n : (\Delta^{op})^n \times \text{Cat}^{pb} \xrightarrow{\text{Tw}(-)^{\times n} \times \text{id}} (\text{Cat}^{op})^{\times n} \times \text{Cat}^{pb} \xrightarrow{(\times^n) \times \text{id}} \text{Cat}^{op} \times \text{Cat}^{pb} \xrightarrow{\text{Fun}(-, -)} \text{Cat} \xrightarrow{(-)^{\simeq}} \text{Spc}$$

On the other hand, noticing the natural projection $\text{Tw}(-) \Rightarrow \text{id}$, we get a subfunctor of $\overline{\text{SPAN}}_n$

$$\overline{\text{SPAN}}_{(n-1)}^+ : (\Delta^{op})^{n-1} \times \Delta^{op} \times \text{Cat}^{pb} \xrightarrow{(\text{Tw}(-)^{\times n-1} \times \text{id}) \times \text{id}} (\text{Cat}^{op})^{\times n} \times \text{Cat}^{pb} \rightarrow \text{Cat}^{op} \times \text{Cat}^{pb} \xrightarrow{\text{Fun}(-, -)} \text{Cat} \rightarrow \text{Spc}$$

Moreover, denote $\Lambda_n \subset \text{Tw}(n)$ as the full subcategory consisting of $(i \leq j)$ with $j - i \leq 1$, by taking the subspace

$$\text{Map}(\text{Tw}(m_1) \times \cdots \times \text{Tw}(m_n), \mathcal{C})^{\text{cart}} \subset \text{Fun}(\text{Tw}(m_1) \times \cdots \times \text{Tw}(m_n), \mathcal{C})^{\simeq}$$

consisting of those functors $F : \text{Tw}(m_1) \times \cdots \times \text{Tw}(m_n) \rightarrow \mathcal{C}$ which are the right Kan extension of $F|_{\Lambda_{m_1} \times \cdots \times \Lambda_{m_n}}$ along $\Lambda_{m_1} \times \cdots \times \Lambda_{m_n} \rightarrow \text{Tw}(m_1) \times \cdots \times \text{Tw}(m_n)$, we get a subfunctor

$$\text{SPAN}_n \hookrightarrow \overline{\text{SPAN}}_n : (\Delta^{op})^n \times \text{Cat}^{pb} \rightarrow \text{Spc}$$

Given $\mathcal{C} \in \text{Cat}^{pb}$, there is an associated multisimplicial space $\text{SPAN}_n(\mathcal{C}) : (\Delta^{op})^n \rightarrow \text{Spc}$, and by [12, proposition 5.14] and [8, proposition 4.9], there is:

Lemma 2.1.2. For $\mathcal{C} \in \text{Cat}^{pb}$, $\text{SPAN}_n(\mathcal{C})$ is an n -fold category object in Spc .

On the other hand, for $\overline{\text{SPAN}}_{n-1}^+$, we acquire a subfunctor:

$$\text{SPAN}_{n-1}^+ \hookrightarrow \overline{\text{SPAN}}_{n-1}^+ : (\Delta^{op})^n \times \text{Cat}^{pb} \rightarrow \text{Spc}$$

by taking the subspace

$$\text{Map}(\text{Tw}(m_1) \times \cdots \times \text{Tw}(m_{n-1}) \times [m_n], \mathcal{C})^{cart} \subset \text{Map}([m_n], \text{Fun}(\text{Tw}(m_1) \times \cdots \times \text{Tw}(m_{n-1}), \mathcal{C}))$$

consisting of those functors $F : \text{Tw}(m_1) \times \cdots \times \text{Tw}(m_{n-1}) \times [m_n] \rightarrow \mathcal{C}$ that are the right Kan extension of $F|_{\Lambda_{m_1} \times \cdots \times \Lambda_{m_{n-1}} \times [m_n]}$ along $\Lambda_{m_1} \times \cdots \times \Lambda_{m_{n-1}} \times [m_n] \rightarrow \text{Tw}(m_1) \times \cdots \times \text{Tw}(m_{n-1}) \times [m_n]$.

And especially by [12, lemma 5.6], there is a subfunctor inclusion $\text{SPAN}_{n-1}^+ \rightarrow \text{SPAN}_n : (\Delta^{op})^n \times \text{Cat}^{pb} \rightarrow \text{Spc}$. For $\mathcal{C} \in \text{Cat}^{pb}$, $\text{SPAN}_+(\mathcal{C}) : (\Delta^{op})^n \rightarrow \text{Spc}$ can be regarded as a functor $(\Delta^{op})^{n-1} \rightarrow \text{Fun}(\Delta^{op}, \text{Spc})$. There is

Lemma 2.1.3. For $\mathcal{C} \in \text{Cat}^{pb}$,

- (1) $\overline{\text{SPAN}}_{n-1}^+(-) : (\Delta^{op})^{n-1} \times \Delta^{op} \times \text{Cat}^{pb} \rightarrow \text{Spc}$ corresponds to a functor $\overline{\text{SPAN}}_{n-1}^+(-)' : (\Delta^{op})^{n-1} \times \text{Cat}^{pb} \rightarrow \text{Cat} \subset \text{Fun}(\Delta^{op}, \text{Spc})$ under the Hom-tensor adjunction, which coincides with the functor

$$(\Delta^{op})^{n-1} \times \text{Cat}^{pb} \xrightarrow{\text{Tw}(-)^{\times n-1} \times \text{id}} (\text{Cat}^{op})^{n-1} \times \text{Cat}^{pb} \xrightarrow{\times^{n-1} \times \text{id}} \text{Cat}^{op} \times \text{Cat}^{pb} \xrightarrow{\text{Fun}(-, -)} \text{Cat}$$

- (2) And $\text{SPAN}_{n-1}^+(-) : (\Delta^{op})^{n-1} \times \Delta^{op} \times \text{Cat}^{pb} \rightarrow \text{Spc}$ corresponds to the subfunctor of $\overline{\text{SPAN}}_{n-1}^+(-)'$ consisting of those functors $F : \text{Tw}(m_1) \times \cdots \times \text{Tw}(m_{n-1}) \rightarrow \mathcal{C}$ which are the right Kan extensions of $F|_{\Lambda_{m_1} \times \cdots \times \Lambda_{m_{n-1}}}$ along $\Lambda_{m_1} \times \cdots \times \Lambda_{m_{n-1}} \hookrightarrow \text{Tw}(m_1) \times \cdots \times \text{Tw}(m_{n-1})$

- (3) For $\mathcal{C} \in \text{Cat}^{pb}$, $\text{SPAN}_{n-1}^+(\mathcal{C})$ is a n -fold category object in Spc and $\text{SPAN}_{n-1}^+(\mathcal{C})([m_1], \dots, [m_{n-1}], -) : \Delta^{op} \rightarrow \text{Spc}$ is a complete Segal space for any $([m_1], \dots, [m_{n-1}]) \in (\Delta^{op})^{n-1}$.

- (4) Moreover, there is a commutative diagram

$$\begin{array}{ccc} (\Delta^{op})^{n-1} \times \text{Cat}^{pb} & \xrightarrow{\text{SPAN}_{n-1}^+(-)'} & \text{Cat} \\ & \searrow \text{SPAN}_{n-1} & \swarrow (-)^{\simeq} \\ & \text{Spc} & \end{array}$$

Proof. (1) and (2) are due to the following facts:

- Noticing the following commutative diagram

$$\begin{array}{ccc} \text{Cat}^{op} \times \text{Cat}^{op} \times \text{Cat} & \xrightarrow{\text{id} \times \text{Fun}(-, -)} & \text{Cat}^{op} \times \text{Cat} \\ \downarrow (\times) \times \text{id} & & \downarrow \text{Fun}(-, -) \\ \text{Cat}^{op} \times \text{Cat} & \xrightarrow{\text{Fun}(-, -)} & \text{Cat} \end{array}$$

- Noticing that the functor

$$\text{Cat} \hookrightarrow \text{Fun}(\Delta^{op}, \text{Spc})$$

corresponds to the functor

$$\Delta^{op} \times \text{Cat} \xrightarrow{\text{Fun}(-, -)} \text{Cat} \xrightarrow{(-)^\approx} \text{Spc}$$

- Thus, the functor

$$(\Delta^{op})^{n-1} \times \text{Cat}^{pb} \xrightarrow{\text{Tw}(-)^{\times n-1} \times \text{id}} (\text{Cat}^{op})^{n-1} \times \text{Cat}^{pb} \xrightarrow{(-)^{n-1} \times \text{id}} \text{Cat}^{op} \times \text{Cat}^{pb} \xrightarrow{\text{Fun}(-, -)} \text{Cat} \hookrightarrow \text{Fun}(\Delta^{op}, \text{Spc})$$

corresponds exactly to the functor

$$\overline{\text{SPAN}}_{n-1}^+ : (\Delta^{op})^{n-1} \times \Delta^{op} \times \text{Cat}^{pb} \xrightarrow{\text{Tw}(-)^{\times n-1} \times \text{id} \times \text{id}} (\text{Cat}^{op})^{\times n-1} \times \text{Cat}^{op} \times \text{Cat}^{pb} \xrightarrow{(\times) \times \text{id}} \text{Cat}^{op} \times \text{Cat}^{pb} \xrightarrow{\text{Fun}(-, -)} \text{Cat} \rightarrow \text{Spc}$$

under the Hom-tensor adjunction.

(3) and (4) are direct consequences of (1) and (2). □

By [12, proposition 4.12], there is a right adjoint functor

$$\mathcal{U}_{\text{Seg}_n} : \text{Cat}_n(\text{Spc}) \rightarrow \text{Seg}_n(\text{Spc})$$

from the category of n -fold category objects in Spc to the category of n -Segal spaces, furthermore, there is a left adjoint functor

$$L : \text{Seg}_n(\text{Spc}) \rightarrow \text{CSS}_n(\text{Spc}) \cong \text{Cat}_n$$

from the category of n -Segal space to the category of complete n -Segal space, which is one model of (∞, n) categories.

What's more, one can show that after taking $\mathcal{U}_{\text{Seg}_n}$, for $\text{SPAN}_n(\mathcal{C})$ and $\text{SPAN}_{n-1}^+(\mathcal{C})$, they have already been the complete n -Segal space:

Lemma 2.1.4. *For $\mathcal{C} \in \text{Cat}^{pb}$, and $n \geq 1$, both $\mathcal{U}_{\text{Seg}_n}(\text{SPAN}_n(\mathcal{C}))$ and $\mathcal{U}_{\text{Seg}_n}(\text{SPAN}_{n-1}^+(\mathcal{C}))$ are complete n -Segal spaces.*

Proof. The case of $\mathcal{U}_{\text{Seg}}(\text{SPAN}_n(\mathcal{C}))$ is by [12, corollary 8.5].

As for $\text{SPAN}_{n-1}(\mathcal{C})$, we follow the same proof strategy:

- By induction, the case when $n = 1$ is trivial.
- And for $n \geq 2$, by the same proof of [12, proposition 8.3], for any $x, y \in \mathcal{C}$ $\mathcal{U}_{\text{Seg}_n} \text{SPAN}_{n-1}^+(\mathcal{C})(x, y) \cong \mathcal{U}_{\text{Seg}_{n-1}} \text{SPAN}_{n-2}^+(\mathcal{C}_{/x \times y})$, which is complete by induction.
- Thus, both $\mathcal{U}_{\text{Seg}_n} \text{SPAN}_{n-1}^+(\mathcal{C})(x, y) \cong \mathcal{U}_{\text{Seg}_{n-1}} \text{SPAN}_{n-2}^+(\mathcal{C}_{/x \times y})$ and $\mathcal{U}_{\text{Seg}_n} \text{SPAN}_{n-1}^+(\mathcal{C})_{\bullet, 0, \dots, 0} \cong \mathcal{U}_{\text{Seg}_1} \text{SPAN}_1(\mathcal{C})$ are complete, and $\mathcal{U}_{\text{Seg}_n}(\text{SPAN}_{n-1}^+(\mathcal{C}))$ is also complete by [12, theorem 7.18]

□

Remark 2.1.5. Especially, the functor $\mathcal{U}_{\text{Seg}_n}$ is just taking sub-multisimplicial space, thus all the inclusion relations for SPAN_{n-1}^+ and SPAN_n still hold.

By composition with $\mathcal{U}_{\text{Seg}_n}$, there are functors

$$\begin{aligned}\text{Span}_n(-) &= \mathcal{U}_{\text{Seg}_n} \circ \text{SPAN}_n : \text{Cat}^{pb} \rightarrow \text{Cat}_n \\ \text{Span}_{n-1}^+(-) &= \mathcal{U}_{\text{Seg}_n} \circ \text{SPAN}_{n-1}^+ : \text{Cat}^{pb} \rightarrow \text{Cat}_n\end{aligned}$$

and there is a natural inclusion $\iota_{n-1} : \text{Span}_{n-1}^+(-) \Rightarrow \text{Span}_n(-)$.

Remark 2.1.6. By the above lemmas, there is a commutative diagram:

$$\begin{array}{ccc} \text{Cat}^{pb} & \xrightarrow{\text{Span}_{n-1}^+(-)} & \text{Cat}_n \\ & \searrow \text{Span}_{n-1}(-) \quad \swarrow \tau_n & \\ & \text{Cat}_{n-1} & \end{array}$$

, by taking the right adjoint of the inclusion-truncation adjunction

$$\tau_n : \text{Cat}_n \rightleftarrows \text{Cat}_{n-1} : i$$

, there is a natural transformation $\iota_+ : \text{Span}_{n-1}(-) \Rightarrow \text{Span}_{n-1}^+(-) : \text{Cat}^{pb} \rightarrow \text{Cat}_n$, which value at $\mathcal{C} \in \text{Cat}^{pb}$ being the inclusion $\text{Span}_{n-1}(\mathcal{C}) \rightarrow \text{Span}_{n-1}^+(\mathcal{C})$, thus ι_{n-1}^+ exhibits $\text{Span}_{n-1}(-)$ as a subfunctor of $\text{Span}_{n-1}^+(-)$.

Corollary 2.1.7. In conclusion, there are inclusions of functors:

$$\text{Span}_{n-1}(-) \xrightarrow{\iota_{n-1}^+} \text{Span}_{n-1}^+(-) \xrightarrow{\iota_{n-1}} \text{Span}_n^+(-)$$

.

There are some identifications we can make in comparison with the construction in [18] and [35]:

Remark 2.1.8. The functor $\text{Span}_1(-) : \text{Cat}^{pb} \rightarrow \text{Cat}$ coincides with the functor $\text{Corr}(-) : \text{GeomSetup} \rightarrow \text{Cat}$ by the embedding $\text{Cat}^{pb} \hookrightarrow \text{GeomSetup}$.

Remark 2.1.9. Noticing that $\overline{\text{SPAN}}_1^+ : \Delta^{op} \times \Delta^{op} \times \text{Cat}^{pb} \rightarrow \text{Spc}$ coincides with the functor in [35, construction 3.2.1]:

$$\overline{2\text{Corr}}(-) : \Delta^{op} \times \text{Cat}^{pb} \xrightarrow{\text{Tw}(-) \times \text{id}} \text{Cat}^{op} \times \text{Cat}^{pb} \xrightarrow{\text{Fun}(-, -)} \text{Cat} \hookrightarrow \text{Fun}(\Delta^{op}, \text{Spc})$$

, thus the functor $2\text{Corr}(-) : \text{Cat}^{pb} \rightarrow \text{Cat}_2$ loc. cit is exactly the functor $\text{Span}_1^+(-)$.

2.2 Symmetric monoidal structure on Span

In this subsection, we use the functoriality of $\text{Span}_n(-)$ to enhance $\text{Span}_n(\mathcal{C})$ into a symmetric monoidal (∞, n) category.

Construction 2.2.1. By [7, lemma 4.1 & remark 4.1], the functors

$$\text{Span}_n(-) : \text{Cat}^{pb} \rightarrow \text{Cat}_n, \text{Span}_{n-1}^+(-) : \text{Cat}^{pb} \rightarrow \text{Cat}_n$$

preserve finite limits; thus, they are enhanced into symmetric monoidal functors

$$\text{Span}_n(-) : (\text{Cat}^{pb})^\times \rightarrow (\text{Cat}_n)^\times, \text{Span}_{n-1}^+ : (\text{Cat}^{pb})^\times \rightarrow (\text{Cat}_n)^\times$$

And for $\mathcal{C} \in \text{Cat}^{pb}$, consider the Cartesian symmetric monoidal structure on $\mathcal{C}$. We get $\mathcal{C} \in \text{CAlg}(\text{Cat}^{pb}) \mapsto \text{Span}_n(\mathcal{C}) \in \text{CAlg}(\text{Cat}_n)$; thus, $\text{Span}_n(\mathcal{C})$ is enhanced into a symmetric monoidal (∞, n) category; so is $\text{Span}_{n-1}^+(\mathcal{C})$.

Moreover, consider the subfunctor $\iota : \text{Span}_1^+(-) \Rightarrow \text{Span}_2(-)$, which is encoded as a functor $\iota : [1] \rightarrow \text{Fun}(\text{Cat}^{pb}, \text{Cat}_2)$ that factors through the full subcategory $\text{Fun}^\times(\text{Cat}^{pb}, \text{Cat}_2) \subset \text{Fun}(\text{Cat}^{pb}, \text{Cat}_2)$ consisting of the finite-product-preserving functors. By [25, proposition 2.4.1.6], $\iota : [1] \rightarrow \text{Fun}^\times(\text{Cat}^{pb}, \text{Cat}_2) \cong \text{Fun}^\otimes((\text{Cat}^{pb})^\times, (\text{Cat}_2)^\times)$, thus $\iota : \text{Span}_{n-1}(\mathcal{C}) \rightarrow \text{Span}_n(\mathcal{C})$ can be enhanced into a symmetric monoidal (∞, n) functor.

$$\iota_{n-1} : \text{Span}_{n-1}^+(\mathcal{C}) \rightarrow \text{Span}_n(\mathcal{C})$$

Likewise, $\iota_{n-1}^+ : \text{Span}_{n-1}(\mathcal{C}) \rightarrow \text{Span}_{n-1}^+(\mathcal{C})$ can also be enhanced into a symmetric monoidal (∞, n) functor.

Remark 2.2.2. A symmetric monoidal (∞, n) category $\mathcal{M}$ is equivalent to a functor $\text{Comm}^\otimes \rightarrow \text{Cat}_n$. By the (∞, n) (un)straightening (for example, constructed in [29]), it corresponds to an (∞, n) coCartesian fibration $\mathcal{M}^\otimes \rightarrow \text{Comm}^\otimes$. Under this correspondence,

$$\iota_{n-1} : \text{Span}_{n-1}^+(\mathcal{C})^\otimes \rightarrow \text{Span}_n(\mathcal{C})^\otimes$$

can be characterized as the non-full (∞, n) subcategory inclusion, where it contains all the coCartesian liftings, and inside a fixed fiber, it contains the n -morphisms in the form of a span, with the left leg being an equivalence. Likewise,

$$\iota_{n-1}^+ : \text{Span}_{n-1}(\mathcal{C})^\otimes \rightarrow \text{Span}_{n-1}^+(\mathcal{C})^\otimes$$

can be characterized as the non-full (∞, n) subcategory inclusion, where it contains all the coCartesian liftings, and inside a fixed fiber, it contains only the invertible n -morphisms. In conclusion, we can regard ι_{n-1} and ι_{n-1}^+ as symmetric monoidal (∞, n) functors which underlying (∞, n) functors are non-full inclusions.

Remark 2.2.3. By remark 2.1.9, we can see that the symmetric monoidal structure on $\text{Span}_1^+(\mathcal{C})$ coincides with the symmetric monoidal structure on $2\text{Corr}(\mathcal{C})$ in [35, remark 3.2.3].

Remark 2.2.4. Moreover, the composition of symmetric monoidal inclusions $\mathcal{C} \xrightarrow{\iota_0} \text{Span}_1(\mathcal{C}) \xrightarrow{\iota_1^+} \text{Span}_1^+(\mathcal{C})$ is equivalent to the symmetric monoidal inclusion $\iota_{\mathcal{C}} : \mathcal{C} \rightarrow 2\text{Corr}(\mathcal{C})$ constructed in [35, Construction 3.2.4]. Both inclusions are characterized as the same non-full $(\infty, 1)$ subcategory of the $(\infty, 2)$ category $\text{Span}_1^+(\mathcal{C})^{\otimes}$ over $\text{Comm}^{\otimes}$, thus satisfying the universal property established in [35, corollary 1.2.2]: a symmetric monoidal $(\infty, 2)$ functor $\mathcal{C} \rightarrow \mathcal{D}$ admits a unique factorization through $\text{Span}_1^+(\mathcal{C})$ if it satisfies the Beck-Chevalley condition .

On the other hand, in [18, section 2.3], there is a symmetric monoidal structure constructed on a geometric setup (C, C_0) :

$$\text{Span}_1(C, C_0)^{\otimes} := \text{Span}_1(C^{op, \Pi})^{op}, (C_0^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq} \rightarrow \text{Comm}^{\otimes}$$

which is an ∞ -operad and is a symmetric monoidal ∞ -category when C admits finite products. And by [18, example 2.3.6], when a morphism between geometric setups $(f, f_0) : (C, C_0) \rightarrow (D, D_0)$ satisfies that f preserves finite products, then the induced morphism $\text{Span}_1(C, C_0)^{\otimes} \rightarrow \text{Span}_1(D, D_0)^{\otimes}$ is symmetric monoidal.

Especially, by taking the geometric setup to be $(\mathcal{C}, \mathcal{C})$, we get another symmetric monoidal structure on $\text{Span}_1(\mathcal{C})$, and in the remaining part, we are going to show that this is equivalent to the one acquired from the functoriality of $\text{Span}_1(-)$.

Remark 2.2.5. • In [18, proposition 2.3.5], the functor $\text{Span}_1(-)^{\otimes} : \text{GeomSetup} \rightarrow \text{Op}$ is constructed in the following way:

$$\begin{array}{ccc} \text{GeomSetup} & \xrightarrow{C \mapsto ((C^{op, \Pi})^{op})} & \text{GeomSetup}_{/(\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, op})} \xrightarrow{\text{Span}_1(-)} \text{Cat}_{/\text{Span}_1(\text{Comm}^{\otimes, op})} \\ & & \text{pullback along } \text{Comm}^{\otimes} \rightarrow \text{Span}_1(\text{Comm}^{\otimes, op}) \xrightarrow{\quad} \text{Cat}_{/\text{Comm}^{\otimes}} \end{array}$$

- Notice that when $C \rightarrow D$ preserves pullback, given $e : d \rightarrow c \in C$, there is a commutative diagram:

$$\begin{array}{ccc} C/c & \longrightarrow & D/F(c) \\ e^* \downarrow & & \downarrow F(e)^* \\ C/d & \longrightarrow & D/F(d) \end{array}$$

- And we apply the above observation to the pullback-preserving functor $\text{Span}_1(-)$, and $e : (\langle 1 \rangle, \langle 1 \rangle) \rightarrow (\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, op})$, then there is a commutative diagram:

$$\begin{array}{ccccc} & & \text{Span}_1(-)^{\otimes} & & \\ & \text{GeomSetup} & \xrightarrow{\quad} & \text{Cat}_{/\text{Span}_1(\text{Comm}^{\otimes, op})} & \xrightarrow{\quad} \text{Op} \subset \text{Cat}_{/\text{Comm}^{\otimes}} \\ & \downarrow (-)^{op, \Pi, op} & & \downarrow \text{Span}_1(e)^* & \\ \text{GeomSetup}_{/(\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, op})} & \xrightarrow{\text{Span}_1(-)} & \text{Cat}_{/\text{Span}_1(\text{Comm}^{\otimes, op})} & \xrightarrow{\quad} & \text{Op} \subset \text{Cat}_{/\text{Comm}^{\otimes}} \\ & \downarrow e^* & & & \\ \text{GeomSetup}_{/*=(\langle 1 \rangle, \langle 1 \rangle)} & \xrightarrow{\text{Span}_1(-)} & \text{Cat}_{/*=\{\langle 1 \rangle\}} & & \end{array}$$

- Thus we have a commutative diagram:

$$\begin{array}{ccccc}
 \text{Cat}^{pb} & \xrightarrow{\text{Span}(-)^{\otimes}} & \text{Cat}^{\otimes} & \hookrightarrow & \text{Op} \\
 & \searrow \text{Span}(-) & \searrow & & \downarrow G \\
 & & & & \text{Cat}
 \end{array}$$

where:

- G is induced by pullback along $* \cong \{1\} \rightarrow \text{Comm}^{\otimes}$ and sends an ∞ -operad to its underlying category.
- Moreover, after precomposition with $\text{Cat}^{pb} \rightarrow \text{GeomSetup}$, the functor assigning the symmetric monoidal structure $\text{Span}_1(-)^{\otimes}$:

$$\text{Span}(-)^{\otimes} : \text{GeomSetup} \rightarrow \text{Op}$$

factors through the non full subcategory of Op consisting of symmetric monoidal ∞ -categories and symmetric monoidal ∞ -functors between them.

- Thus, there is a factorization

$$\text{Span}_1(-)^{\otimes} : \text{Cat}^{pb} \rightarrow \text{Cat}^{\otimes} \subset \text{Op}$$

.

Moreover, there is:

Lemma 2.2.6. *The above functor $\text{Cat}^{pb} \rightarrow \text{GeomSetup} \xrightarrow{\text{Span}_1(-)^{\otimes}} \text{Cat}^{\otimes}$, $C \mapsto (C, C) \mapsto \text{Span}_1(C)^{\otimes}$ preserves finite products.*

Proof. • Obviously $\text{Span}_1(*)^{\otimes} \cong \text{Span}_1(\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, \simeq}) \cong \text{Comm}^{\otimes}$, thus $\text{Span}_1(-)^{\otimes}$ preserves the final object. And it remains to show that it preserves products.

- Recall the construction of the coCartesian symmetric monoidal structure in [25, Construction 2.4.3.1]. Given an ∞ category C , C^{II} is defined as

$$\text{Hom}_{\text{N}(\text{Fin})_*}(K, C^{\text{II}}) \cong \text{Hom}_{\text{sSet}}(K \times_{\text{N}(\text{Fin})_*} \Gamma^*, C)$$

. it's obvious that for $C \times D$, the representator on the left hand side should be $C^{\text{II}} \times D^{\text{II}}$, which implies $(C \times D)^{\text{II}} \cong (C^{\text{II}} \times_{\text{Comm}^{\otimes}} D^{\text{II}})$, where the second product takes place in $\text{Cat}_{/\text{N}(\text{Fin})_*}$. Thus, it's the fiber product over $\text{N}(\text{Fin})_*$.

- So there are equivalences in Cat :

$$((C \times D)^{op, \text{II}})^{op} \cong (C^{op, \text{II}} \times_{\text{Comm}^{\otimes}} D^{op, \text{II}})^{op} \cong (C^{op, \text{II}})^{op} \times_{\text{Comm}^{\otimes, op}} (D^{op, \text{II}})^{op}$$

, and so are

$$((C \times D)^{op, \text{II}})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq} \cong ((C^{op, \text{II}})^{op} \times_{\text{Comm}^{\otimes, op}} (D^{op, \text{II}})^{op}) \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}$$

and

$$\begin{aligned} & ((C^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} (D^{op, \Pi})^{op}) \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq} \cong \\ & ((C^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}) \times_{\text{Comm}^{\otimes, op}} ((D^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}) \end{aligned}$$

by the fact that $\text{Comm}^{\otimes, \simeq} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq} \cong \text{Comm}^{\otimes, \simeq}$ and limit commutes with limit.

- Thus we have a fiber product in GeomSetup:

$$\begin{array}{ccc} (((C \times D)^{op, \Pi})^{op}, ((C \times D)^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}) & \xrightarrow{\quad} & ((C^{op, \Pi})^{op}, (C^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}) \\ \downarrow \lrcorner & & \downarrow \\ ((D^{op, \Pi})^{op}, (D^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}) & \xrightarrow{\quad} & (\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, \simeq}) \end{array}$$

•

- Since $\text{Span}_1(-) : \text{GeomSetup} \rightarrow \text{Cat}$ preserves finite limits, thus there is a pullback diagram in Cat:

$$\begin{array}{ccc} \text{Span}_1(C \times D)^{\otimes} & \xrightarrow{\quad} & \text{Span}_1(C)^{\otimes} \\ \downarrow \lrcorner & & \downarrow \\ \text{Span}_1(D)^{\otimes} & \xrightarrow{\quad} & \text{Comm}^{\otimes} \end{array}$$

here we identify $\text{Span}_1(\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, \simeq}) \cong \text{Comm}^{\otimes}$ by [18, example 2.2.13].

- This implies we have a product in $\text{Cat}^{\otimes} \subset \text{Op}$:

$$\text{Span}_1(C \times D)^{\otimes} \cong \text{Span}_1(C)^{\otimes} \times_{\text{Comm}^{\otimes}} \text{Span}_1(D)^{\otimes}$$

- Thus $\text{Span}_1(-)^{\otimes} : \text{Cat}^{pb} \rightarrow \text{Cat}^{\otimes}$ preserves products.

□

Thus $\text{Cat}^{pb} \xrightarrow{\text{Span}(-)^{\otimes}} \text{Cat}^{\otimes}$ preserves products, and so is the functor G by [25, remark 2.1.4.10], thus there is a commutative diagram:

$$\begin{array}{ccc} \text{Cat}^{pb, \times} & \xrightarrow{\text{Span}(-)^{\otimes}} & (\text{Cat}^{\otimes})^{\times} \cong \text{Mon}_{\text{Comm}}(\text{Cat})^{\times} \\ & \searrow \text{Span}(-) & \downarrow G \\ & & \text{Cat}^{\times} \end{array}$$

which induces the following commutative diagram:

$$\begin{array}{ccccc}
\text{Cat}^{pb} & \xrightarrow{\text{Span}_1(-)^\otimes} & & & \\
\uparrow \text{ev}_{(1)} & & & & \\
\text{CAlg}(\text{Cat}^{pb}) & \xrightarrow{\text{CAlg}(\text{Span}_1(-)^\otimes)} & \text{CAlg}(\text{Mon}_{\text{Comm}}(\text{Cat})) \cong \text{CAlg}(\text{CAlg}(\text{Cat})) & \xrightarrow{\text{ev}_{(1)}, \cong} & \text{Mon}_{\text{Comm}}(\text{Cat}) \\
& \searrow \text{CAlg}(\text{Span}_1(-)) & \downarrow \cong, \text{CAlg}(G) & & \downarrow G \\
& & \text{CAlg}(\text{Cat}) & \xrightarrow{\text{ev}_{(1)}} & \text{Cat}
\end{array}$$

Thus, given a commutative algebra object $\mathcal{C}$ in Cat^{pb} , the induced commutative algebra object $\mathcal{C} \xrightarrow{*} \text{CAlg}(\text{Cat}^{pb}) \xrightarrow{\text{Span}_1(-)} \text{CAlg}(\text{Cat})$ corresponds to one symmetric monoidal structure on $\text{Span}_1(\mathcal{C})$. On the other hand, if we forget the commutative algebra structure on $\mathcal{C}$ and regard it as an object in Cat^{pb} , by the construction $\text{Span}_1(-)^\otimes$, there is another symmetric monoidal structure on $\text{Span}_1(\mathcal{C})$. These symmetric monoidal structures coincide since $\text{ev}_{(1)} : \text{CAlg}(\text{CAlg}(\text{Cat})) \rightarrow \text{CAlg}(\text{Cat})$ is an equivalence.

2.3 $B : \text{Alg}(\text{Cat}_{n-1}) \rightleftharpoons (\text{Cat}_n)_{*/} : \Omega$

In this part, we review the symmetric monoidal adjunction

$$B : \text{Alg}(\text{Cat}_{n-1}) \rightleftharpoons \text{Cat}_{n*/} : \Omega$$

established in [12, remark 10.6] and [10, theorem 6.3.2], we show that $\Omega(\text{Span}_n(\mathcal{C})) \cong \text{Span}_{n-1}(\mathcal{C})$, and the induced symmetric monoidal structure on $\Omega(\text{Span}_2(\mathcal{C})) \cong \text{Span}_1(\mathcal{C})$ coincides with the symmetric monoidal structure on $\text{Span}_1(\mathcal{C})$ in [construction 2.2.1](#).

Notation 2.3.1. Let $\text{Seg}(\text{Cat}_{n-1})$ denote the full subcategory of $\text{Cat}(\text{Cat}_{n-1})$ consisting of those functors $F : \Delta^{op} \rightarrow \text{Cat}_{n-1}$ satisfying $F([0]) \in \text{Spc} \subset \text{Cat}_{n-1}$.

Using the Segal category model, there is a functor

$$(*) \quad \text{Alg}(\text{Cat}_{n-1}) \hookrightarrow \text{Seg}(\text{Cat}_{n-1}) \xrightarrow{L} \text{Cat}_n$$

where L is the localization functor defined in [12, theorem 7.6].

On the other hand, in the model of (∞, n) category by enriched categories using categorical algebras, there is a functor defined in [10, definition 6.3.1]:

$$(**) \quad \text{Alg}(\text{Cat}_{n-1}) \hookrightarrow \text{Alg}_{\text{cat}}(\text{Cat}_{n-1}) \rightarrow \text{Cat}^{\text{Cat}_{n-1}}$$

where the second functor is the localization functor.

By the comparison of the two models, established in [11], there is a commutative diagram:

$$\begin{array}{ccc}
\text{Cat}^{\text{Cat}_{n-1}} & \hookrightarrow & \text{Alg}_{\text{cat}}(\text{Cat}_{n-1}) \\
\uparrow \cong & & \uparrow \cong \\
\text{Cat}_n & \hookrightarrow & \text{Seg}(\text{Cat}_{n-1})
\end{array}$$

, and by taking the left adjoints, we get

$$\begin{array}{ccc}
\text{Cat}^{\text{Cat}_{n-1}} & \xleftarrow{\text{localization}} & \text{Alg}_{\text{cat}}(\text{Cat}_{n-1}) \\
\downarrow \cong & & \downarrow \cong \\
\text{Cat}_n & \xleftarrow{\text{localization}} & \text{Seg}(\text{Cat}_{n-1})
\end{array}$$

under the equivalence of $\text{Cat}^{\text{Cat}_{n-1}} \cong \text{Cat}_n$, the above two functors (*) and (**) from $\text{Alg}(\text{Cat}_{n-1}) \rightarrow (\text{Cat}_n)_{*/}$ coincide, and we denote it as

$$B : \text{Alg}(\text{Cat}_n) \rightarrow (\text{Cat}_n)_{*/}$$

Lemma 2.3.2. (1) *B is fully faithful.*

(2) *B is a symmetric monoidal functor, with $\text{Alg}(\text{Cat}_n)$ and $(\text{Cat}_n)_{*/} \cong \text{Alg}_{\mathbb{E}_0}(\text{Cat}_n)$ equipped with both the Cartesian symmetric monoidal structure.*

(3) *The functor B admits a right adjoint Ω .*

Proof. This is [10, theorem 6.3.2], combined with the fact that the symmetric monoidal structure on $\text{Cat}^{\text{Cat}_{n-1}}$ is Cartesian by [36, proposition 2.5.8] $\square$

Remark 2.3.3. Moreover, one can describe the right adjoint Ω explicitly: Consider $\text{Seg}(\text{Cat}_{n-1}) \cong \text{Alg}_{\text{cat}}(\text{Cat}_{n-1})$:

- There is a commutative diagram

$$\begin{array}{ccc}
\text{Seg}(\text{Cat}_{n-1}) & \xrightarrow{\cong} & \text{Alg}_{\text{cat}}(\text{Cat}_{n-1}) \\
\searrow \text{ev}_{[0]} & & \swarrow p \\
& \text{Spc} &
\end{array}$$

where $\text{ev}_{[0]}$ is induced by $[0] \rightarrow \Delta^{op}$, and p is the functor taking the underlying space S of a categorical algebra $\Delta_S^{op} \rightarrow \text{Cat}_{n-1}^\times$. Moreover, the above equivalence $\text{Seg}(\text{Cat}_{n-1}) \rightarrow \text{Alg}_{\text{cat}}(\text{Cat}_{n-1})$ is a morphism between Cartesian fibrations over Spc .

- By [12, lemma 6.33 & theorem 6.3.2], the above adjunction $B \dashv \Omega$ corresponds to the biCartesian fibration

$$\begin{array}{ccc}
\Box & \longrightarrow & \text{Seg}(\text{Cat}_{n-1})_{*/} \\
\downarrow \lrcorner & & \downarrow \\
\text{Spc}_{*/}^{\triangleleft} & \longrightarrow & \text{Spc}_{*/} \\
\downarrow & & \\
[1] & &
\end{array}$$

- Thus $\Omega(X)$ computes the Cartesian lifting of X over $* \rightarrow X([0])$ for $X \in \text{Seg}(\text{Cat}_{n-1})$.

- Noticing that $ev_{[0]} : \text{Seg}(\text{Cat}_{n-1}) \subset \text{Fun}(\Delta^{op}, \text{Cat}_{n-1}) \rightarrow \text{Cat}_{n-1}$ admits a right adjoint R by taking the Čech nerve of an object to the final object, then by [13, corollary 4.5.2], we can compute the Cartesian lifting of X over the pointed map $* \rightarrow X[0]$ through the pullback diagram

$$\begin{array}{ccc} \Omega(X) & \xrightarrow{\quad} & X \\ \downarrow & \lrcorner & \downarrow \eta \\ \text{Rev}_{[0]}(*) & \xrightarrow{\quad} & \text{Rev}_{[0]}(X) = X([0])^{\times \bullet + 1} \end{array}$$

where η is the coevaluation map.

- So the underlying object $\Omega(X)[1]$ of $\Omega(X)$ is equivalent to the pullback $X[1] \times_{X([0])^{\times 2}} *$.
- The algebra object structure on $\Omega(X)$ is induced by the composition law for $X([1])$: for the morphism $0, 2 : [1] \rightarrow [2] \in \Delta$, there is a composition map $X([2]) \cong X([1]) \times_{X([0])} X([1]) \rightarrow X([1])$ that restricts to the multiplication map for the algebra object $\Omega(X)[1] \in \text{Cat}_{n-1}$.
- The functoriality of the right adjoint Ω can also be described using the Cartesian lifting: given $f : X \rightarrow Y$, the induced morphism between $\Omega(f) : \Omega(X) \rightarrow \Omega(Y)$ is induced by the pullbacks:

$$\begin{array}{ccccc} \Omega(Y) & \xrightarrow{\quad} & Y & & \\ \downarrow & \searrow & \downarrow & \searrow f & \\ \text{Rev}_{[0]}(*) & \xrightarrow{\quad} & \text{Rev}_{[0]}(Y) & & \\ & \searrow \Omega(f) & \downarrow & \searrow & \\ & & \Omega(X) & \xrightarrow{\quad} & X \\ & & \downarrow & \lrcorner & \downarrow \eta \\ & & \text{Rev}_{[0]}(*) & \xrightarrow{\quad} & \text{Rev}_{[0]}(X) = X([0])^{\times \bullet + 1} \end{array}$$

Lemma 2.3.4. For $\mathcal{C} \in \text{Cat}^{pb}$,

- (1) For, $n \geq 1$, consider the pointed map $* \rightarrow \text{Span}_n(\mathcal{C})$ specifying the final object $* \in \mathcal{C}$, $\Omega(\text{Span}_n(\mathcal{C})) \cong \text{Span}_{n-1}(\mathcal{C})$
- (2) For $n \geq 2$, consider the pointed map $* \rightarrow \text{Span}_{n-1}^+(\mathcal{C})$ specifying the final object $* \in \mathcal{C}$, $\Omega(\text{Span}_{n-1}^+(\mathcal{C})) \cong \text{Span}_{n-2}^+(\mathcal{C})$.

Proof. (1) is by [12, proposition 8.3], and (2) is the same by replacing $\overline{\text{SPAN}}_n$ with $\overline{\text{SPAN}}_{n-1}^+$, SPAN_n with SPAN_{n-1}^+ , and Span_n with Span_{n-1}^+ . $\square$

Noticing that the left adjoint B in the adjunction $B : \text{Alg}(\text{Cat}_{n-1}) \rightleftharpoons (\text{Cat}_n)_{*/} : \Omega$ is symmetric monoidal, thus Ω can be enhanced into a lax symmetric monoidal functor. Given the symmetric monoidal $(\infty, 2)$ category $\text{Span}_2(\mathcal{C})$, the pointed map $* \rightarrow \text{Span}_2(\mathcal{C})$ specifying the final object $* \in \mathcal{C}$ is the unit map, thus it defines an object in $\text{CAlg}((\text{Cat}_n)_{*/})$ and induces a symmetric monoidal structure on $\Omega(\text{Span}_2(\mathcal{C}) \cong \text{Span}_1(\mathcal{C}))$.

To distinguish this symmetric monoidal structure from the one acquired in the last subsection (which is through the functoriality of the functor $\text{Span}_1(-)$), we denote the symmetric monoidal category $\text{Span}_1(\mathcal{C})^{\otimes'}$ as the one from the lax symmetric monoidal functor Ω , and $\text{Span}_1(\mathcal{C})^{\otimes}$ as the one from the functoriality of the finite-limit preserving functor $\text{Span}_1(-) : \text{Cat}^{bb} \rightarrow \text{Cat}_1$. Likewise, let $\text{Span}_1^+(\mathcal{C})^{\otimes'}$ denote the symmetric monoidal structure on $\Omega(\text{Span}_2^+(\mathcal{C}))$ acquired from the lax symmetric monoidal functor Ω , and $\text{Span}_1^+(\mathcal{C})^{\otimes}$ as the one from the functoriality of the functor $\text{Span}_1^+(-)$.

Proposition 2.3.5. *There is a symmetric monoidal equivalence of symmetric monoidal categories $\text{Span}_1(\mathcal{C})^{\otimes} \cong \text{Span}_1(\mathcal{C})^{\otimes'}$ and $\text{Span}_1^+(\mathcal{C})^{\otimes} \cong \text{Span}_1^+(\mathcal{C})^{\otimes'}$.*

Proof. • Input the following morphisms in $\text{CAlg}((\text{Cat}_3)_{*/})$

$$\text{Span}_1^+(\mathcal{C})^{\otimes} \rightarrow \text{Span}_2(\mathcal{C})^{\otimes} \rightarrow \text{Span}_2^+(\mathcal{C})^{\otimes}$$

. By applying the lax symmetric monoidal functor $\Omega : (\text{Cat}_3)_{*/} \rightarrow \text{Alg}(\text{Cat}_2)$, there is an induced chain of morphisms in $\text{CAlg}(\text{Cat}_2)$:

$$\mathcal{C}^{\otimes'} \xrightarrow{i} \text{Span}_1(\mathcal{C})^{\otimes'} \xrightarrow{i^+} \text{Span}_1^+(\mathcal{C})^{\otimes'}$$

.

- $\mathcal{C}^{\otimes'}$ is equipped with the Cartesian symmetric monoidal structure, noting that the algebra object structure on $\mathcal{C}' = \Omega(\text{Span}_1^+(\mathcal{C})) \cong \underline{\text{End}}_{\text{Span}_1^+(\mathcal{C})}(\ast)$ is by composition. And the final object $\ast$ corresponds to the identity map $\ast = \ast = \ast$. Given $\ast \leftarrow x \rightarrow \ast, \ast \leftarrow y \rightarrow \ast \in \Omega(\text{Span}_1^+(\mathcal{C}))$, the composition map gives us $x \circ y = (\ast \leftarrow x \times y \rightarrow \ast)$, and the induced morphisms to $x \circ \ast = x$ and $\ast \circ y = y$ form a product diagram in $\mathcal{C} = \Omega \text{Span}_1^+(\mathcal{C})$.
- Thus, $\mathcal{C}^{\otimes'} \cong \mathcal{C}^{\otimes}$ are equivalent as symmetric monoidal categories, both with Cartesian symmetric monoidal structures.
- By the [remark 2.2.4](#), the inclusion $\iota_2^+ \circ \iota_1 : \mathcal{C} \rightarrow \text{Span}_1^+(\mathcal{C})$ has the universal property established in [\[35, corollary 1.2.2\]](#); thus, there is an induced morphism $\psi : \text{Span}_1^+(\mathcal{C})^{\otimes} \rightarrow \text{Span}_1^+(\mathcal{C})^{\otimes'}$ making the following diagram commute in $\text{CAlg}(\text{Cat}_2)$:

$$\begin{array}{ccccc} \mathcal{C}^{\otimes} & \xrightarrow{i} & \text{Span}_1(\mathcal{C})^{\otimes'} & \xrightarrow{i^+} & \text{Span}_1^+(\mathcal{C})^{\otimes'} \\ \downarrow \iota_0 & & & \nearrow \psi & \\ \text{Span}_1(\mathcal{C})^{\otimes} & & & & \\ \downarrow \iota_1^+ & & & & \\ \text{Span}_1^+(\mathcal{C})^{\otimes} & & & & \end{array}$$

- By the descriptions of the functor $\Omega(-)$ at objects and morphisms in the [remark 2.3.3](#), we can see that at the level of underlying $(\infty, 2)$ categories, the above inclusion $i^+ \circ i : \mathcal{C}^{\otimes} \rightarrow \text{Span}_1^+(\mathcal{C})^{\otimes'}$ is equivalent to the inclusion $\iota_1^+ \iota_0$, which corresponds to the non-full

subcategory of $\text{Span}_1^+(\mathcal{C})$ consisting of all the objects and 1-morphisms in the form of a span with the left leg being an equivalence and invertible 2-morphisms; thus, it also has the same universal property, and this induces an inverse functor ψ^{-1} of ψ at the level of underlying $(\infty, 2)$ categories; thus, ψ is a symmetric monoidal $(\infty, 2)$ equivalence.

- Since $i^+ : \text{Span}_1(\mathcal{C})^{\otimes'} \rightarrow \text{Span}_1^+(\mathcal{C})^{\otimes'}$ exhibits $\text{Span}_1(\mathcal{C})^{\otimes'}$ as the underlying $(\infty, 1)$ -category of $\text{Span}_1^+(\mathcal{C})^{\otimes'}$, the map $\psi \circ i_1^+ : \text{Span}_1(\mathcal{C})^{\otimes} \rightarrow \text{Span}_1^+(\mathcal{C})^{\otimes'}$ factors through $i^+ : \text{Span}_1(\mathcal{C})^{\otimes'} \rightarrow \text{Span}_1^+(\mathcal{C})^{\otimes'}$ and the factorization $\text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_1(\mathcal{C})^{\otimes'}$ is an equivalence at the level of underlying $(\infty, 1)$ categories; thus, it is a symmetric monoidal equivalence.

□

Corollary 2.3.6. *Consider the composition of adjunctions:*

$$B : \text{CAlg}(\text{Cat}) \rightleftharpoons \text{CAlg}(\text{Cat}_2) : \Omega, B : \text{CAlg}(\text{Cat}_2) \rightleftharpoons \text{CAlg}(\text{Cat}_3) : \Omega, \dots, B : \text{CAlg}(\text{Cat}_{n-1}) \rightleftharpoons \text{CAlg}(\text{Cat}_n) : \Omega$$

, then for $\text{Span}_n(\mathcal{C}) \in \text{CAlg}(\text{Cat}_n)$, $\Omega^{n-1}(\text{Span}_n(\mathcal{C})) \cong \text{Span}_1(\mathcal{C})$ in $\text{CAlg}(\text{Cat})$.

Proof. • Again we consider $\text{Span}_{n-1}^+(\mathcal{C}) \rightarrow \text{Span}_n(\mathcal{C}) \rightarrow \text{Span}_n^+(\mathcal{C})$ in $\text{CAlg}(\text{Cat}_{n+1})$, then by applying $\Omega^{n-1}(-)$, we get

$$\mathcal{C}' \rightarrow \text{Span}_1(\mathcal{C})' \rightarrow \text{Span}_1^+(\mathcal{C})'$$

where $\mathcal{C}'$, $\text{Span}_1(\mathcal{C})'$ and $\text{Span}_1^+(\mathcal{C})'$ are the same as $\mathcal{C}$, $\text{Span}_1(\mathcal{C})$ and $\text{Span}_1^+(\mathcal{C})$ at the level of underlying $(\infty, 2)$ categories but with possibly different symmetric monoidal structures induced by the lax symmetric monoidal functor $\Omega^{n-1}(-)$.

- Again one can check that $\mathcal{C}'$ is equipped with the Cartesian symmetric monoidal structure, and thus by the universal property of $\mathcal{C} \rightarrow \text{Span}_1^+(\mathcal{C})$, there is a symmetric monoidal $(\infty, 2)$ equivalence $\psi : \text{Span}_1^+(\mathcal{C}) \rightarrow \text{Span}_1^+(\mathcal{C})'$ which restricts to a symmetric monoidal $(\infty, 1)$ equivalence $\text{Span}_1(\mathcal{C}) \cong \text{Span}_1(\mathcal{C})' = \Omega^{n-1}(\text{Span}_n(\mathcal{C}))$.

□

2.4 $\text{LMod}_{B_A}(\text{Cat}^{\mathcal{M}}) \cong \text{Cat}^{\text{LMod}_A(\mathcal{M})}$

In this subsection, we review the equivalence $\text{Cat}^{\text{LMod}_A(\mathcal{M})} \cong \text{LMod}_{B_A}(\text{Cat}^{\mathcal{M}})$ established in [37, proposition 2.2].

Given a presentably symmetric monoidal ∞ -category $\mathcal{M}$, consider $A \in \text{CAlg}(\mathcal{M})$, then there is a symmetric monoidal adjunction

$$L = (-) \otimes A : \mathcal{M} \rightleftharpoons \text{LMod}_A(\mathcal{M}) : R = \text{forget}$$

.

By [10, proposition 5.7.1], there is a functor

$$\text{Cat}^{(-)} : \text{Alg}(\text{Cat}) \rightarrow \widehat{\text{Cat}}$$

classified by the coCartesian fibration $\text{Enr} \rightarrow \text{Alg}(\text{Cat})$, which is a subfunctor of the functor

$$\text{Alg}_{\text{cat}}(-) : \text{Alg}(\text{Cat}) \rightarrow \widehat{\text{Cat}}$$

,classified by the coCartesian fibration $\text{Alg}_{\text{cat}} \rightarrow \text{Alg}(\text{Cat})$. Moreover, there is an adjunction over $\text{Alg}(\text{Cat})$:

$$\text{loc} : \text{Alg}_{\text{cat}} \rightleftarrows \text{Enr} : i$$

where loc is the localization functor taking a categorical algebra in $\mathcal{V} \in \text{Alg}(\text{Cat})$ to its completion as an $\mathcal{V}$ -enriched category.

By [15, remark 7.11], one can enhance $\text{Enr} \subset \text{Alg}_{\text{cat}} \rightarrow \text{Alg}(\text{Cat})$ into a symmetric monoidal coCartesian fibration $\text{Enr}^\times \subset \text{Alg}^\times \rightarrow \text{Alg}(\text{Cat})^\times$. So one can deduce that:

- The functor $\text{Cat}^{(-)} : \text{Alg}(\text{Cat}) \rightarrow \widehat{\text{Cat}}$ can be enhanced into a lax symmetric monoidal functor. Especially, given a symmetric monoidal functor $L : \mathcal{M} \rightarrow \text{LMod}_A(\mathcal{M})$, $L_* : \text{Cat}^\mathcal{M} \rightarrow \text{Cat}^{\text{LMod}_A(\mathcal{M})}$ is also symmetric monoidal. Thus, there is an induced symmetric monoidal adjunction by [10, proposition 5.7.17]

$$L_* : \text{Cat}^\mathcal{M} \rightleftarrows \text{Cat}^{\text{LMod}_A(\mathcal{M})} : R_*$$

,

- Given $\mathcal{V} \in \text{CAlg}(\text{Cat})$, the localization adjunction

$$\text{loc}_\mathcal{V} : \text{Alg}_{\text{cat}}(\mathcal{V}) \rightleftarrows \text{Cat}^\mathcal{V} : i_\mathcal{V}$$

can be enhanced into a symmetric monoidal adjunction, especially, for any $x \in \text{CAlg}(\text{Alg}_{\text{cat}}(\mathcal{V}))$, the local equivalence $x \rightarrow \text{loc}_\mathcal{V}(x)$ lands in $\text{CAlg}(\text{Alg}_{\text{cat}}(\mathcal{M}))$.

Lemma 2.4.1. *Given a symmetric monoidal adjunction*

$$L : \mathcal{M} \rightleftarrows \mathcal{N} : R$$

, consider the induced symmetric monoidal adjunction

$$L_* : \text{Cat}^\mathcal{M} \rightleftarrows \text{Cat}^\mathcal{N} : R_*$$

, then there is an equivalence $R_*1_\mathcal{N} \cong BR(1_\mathcal{M})$ in $\text{CAlg}(\text{Cat}^\mathcal{N})$ where $BR(1_\mathcal{N})$ is the image of $R(1_\mathcal{N}) \in \text{CAlg}(\mathcal{M})$ under the loop functor $B : \text{CAlg}(\text{Alg}(\mathcal{M})) \rightarrow \text{CAlg}(\text{Cat}^\mathcal{M})$.

Proof. • By definition, the symmetric monoidal functor B is the composition of the following symmetric monoidal functors:

$$B : \text{Alg}(\mathcal{N}) \hookrightarrow \text{Alg}_{\text{cat}}(\mathcal{N})_{1_\mathcal{N}/} \xrightarrow{L_\mathcal{N}} \text{Cat}_{1_{\text{Cat}^\mathcal{N}}/}^\mathcal{N}$$

where $1_\mathcal{N}$ is the unit in $\text{Alg}(\mathcal{N}) \hookrightarrow \text{Alg}_{\text{cat}}(\mathcal{N})$ and $1_{\text{Cat}^\mathcal{N}} = \text{loc}_\mathcal{N}(1_\mathcal{N})$ is its localization (they may differ when the unit $1_\mathcal{N}$ has a nontrivial endomorphism space).

- For $a \in \text{CAlg}(\text{Alg}(\mathcal{N}))$, denote its image in $\text{Alg}_{\text{cat}}(\mathcal{N})_{1_\mathcal{N}/}$ as $B^{\text{pre}} a$. Then, there is a morphism $B^{\text{pre}} 1_\mathcal{N} \rightarrow B 1_\mathcal{N}$ in $\text{CAlg}(\text{Alg}_{\text{cat}}(\mathcal{N}))$ by the symmetric monoidal adjunction $\text{loc}_\mathcal{N} \dashv i_\mathcal{N}$.

- By taking the image of $B^{pre} 1_{\mathcal{N}} \rightarrow B 1_{\mathcal{N}}$ under the functor $\text{CAlg}(R_*) : \text{CAlg}(\text{Alg}_{cat}(\mathcal{N})) \rightarrow \text{CAlg}(\text{Alg}_{cat}(\mathcal{M}))$, we then get a morphism

$$R_* B^{pre} 1_{\mathcal{N}} \cong B^{pre} R(1_{\mathcal{N}}) \rightarrow R_* B 1_{\mathcal{N}}$$

- Especially, by [10, proposition 5.7.17], $R_* B 1_{\mathcal{N}}$ is complete, and the above morphism is fully faithful and essentially surjective by [10, lemma 5.7.5].
- Thus, by the symmetric monoidal adjunction of $L_{\mathcal{M}} \dashv i_{\mathcal{M}}$, the induced morphism $B R(1_{\mathcal{N}}) \cong \text{loc}_{\mathcal{M}}(B^{pre} R(1_{\mathcal{N}})) \rightarrow R_* B 1_{\mathcal{N}}$ is an equivalence in $\text{CAlg}(\text{Cat}^{\mathcal{M}})$.

□

Now consider the lax symmetric monoidal right adjoint $R_* : \text{Cat}^{\text{LMod}_A(\mathcal{M})} \rightarrow \text{Cat}^{\mathcal{M}}$, there is an induced lax symmetric monoidal functor

$$R_* : \text{Cat}^{\text{LMod}_A(\mathcal{M})} \rightarrow \text{LMod}_{R_* 1_{\text{LMod}_A(\mathcal{M})}}(\text{Cat}^{\mathcal{M}}) \cong \text{LMod}_{B A}(\text{Cat}^{\mathcal{M}})$$

where $B A$ is the image of $A \in \text{CAlg}(\mathcal{M})$ under the symmetric monoidal functor $B : \text{Alg}(\mathcal{M}) \rightarrow (\text{Cat}^{\mathcal{M}})_*/$

Moreover, one can show that

Corollary 2.4.2. *In the above context, $R_* : \text{Cat}^{\text{LMod}_A(\mathcal{M})} \rightarrow \text{LMod}_{B A}(\text{Cat}^{\mathcal{M}})$ is a symmetric monoidal equivalence.*

Proof. This is by [37, proposition 2.2] □

Remark 2.4.3. For further purpose, one can show that the above symmetric monoidal equivalence is functorial:

- Given a morphism between commutative algebras $f : A \rightarrow B$ in $\text{CAlg}(\mathcal{M})$, there is a

$$\begin{array}{ccc} \text{LMod}_B(\mathcal{M}) & \xleftarrow{(-) \otimes B} & \mathcal{M} \\ \uparrow (-) \otimes_{AB} & & \parallel \\ \text{LMod}_A(\mathcal{M}) & \xleftarrow{(-) \otimes A} & \mathcal{M} \end{array}, \text{ which gives rise to a}$$

$$\begin{array}{ccc} \text{Cat}^{\text{LMod}_B(\mathcal{M})} & \xleftarrow{((-) \otimes B)_*} & \text{Cat}^{\mathcal{M}} \\ \uparrow ((-) \otimes_{AB})_* & & \parallel \\ \text{Cat}^{\text{LMod}_A(\mathcal{M})} & \xleftarrow{((-) \otimes A)_*} & \text{Cat}^{\mathcal{M}} \end{array}.$$

- By passing to the right adjoints, we get a morphism in Op :

$$\begin{array}{ccc} \text{Cat}^{\text{LMod}_B(\mathcal{M})} & \xrightarrow{(F_{B/1})_*} & \text{Cat}^{\mathcal{M}} \\ (F_{B/A})_* \downarrow & & \parallel \\ \text{Cat}^{\text{LMod}_A(\mathcal{M})} & \xrightarrow{(F_{A/1})_*} & \text{Cat}^{\mathcal{M}} \end{array}$$

where $(F_{B/A})_* : \text{Cat}^{\text{LMod}_B(\mathcal{M})} \rightarrow \text{Cat}^{\mathcal{M}}$ induced by the transfer of enrichment along the forgetful functor $F_{B/A} : \text{LMod}_B(\mathcal{M}) \rightarrow \mathcal{M}$, likewise for $F_{B/1}$ and $F_{A/1}$.

- By taking the induced morphisms between left module categories of the above diagram, we get:

$$\begin{array}{ccc}
\mathrm{LMod}_{1_{\mathrm{Cat}^{\mathrm{LMod}_B(\mathcal{M})}}}(\mathrm{Cat}^{\mathrm{LMod}_B(\mathcal{M})}) \cong \mathrm{Cat}^{\mathrm{LMod}_B(\mathcal{M})} & \xrightarrow{(F_{B/1})_*} & \mathrm{LMod}_B(\mathrm{Cat}^{\mathcal{M}}) \\
\downarrow (F_{B/A})_* & & \downarrow \cong \\
\mathrm{LMod}_B(\mathrm{Cat}^{\mathrm{LMod}_A(\mathcal{M})}) & \xrightarrow{(F_{A/1})_*} & \mathrm{LMod}_B(\mathrm{Cat}^{\mathcal{M}})
\end{array}$$

- Now consider the symmetric monoidal equivalence in [corollary 2.4.2](#): $\mathrm{Cat}^{\mathrm{LMod}_A(\mathcal{M})} \rightarrow \mathrm{LMod}_{B_A}(\mathrm{Cat}^{\mathcal{M}})$, by taking the induced left module categories, there is a commutative diagram:

$$\begin{array}{ccc}
\mathrm{LMod}_B(\mathrm{Cat}^{\mathrm{LMod}_A(\mathcal{M})}) & \xrightarrow{(F_{A/1})_*} & \mathrm{LMod}_B(\mathrm{LMod}_{B_A}(\mathrm{Cat}^{\mathcal{M}})) \cong \mathrm{LMod}_B(\mathrm{Cat}^{\mathcal{M}}) \\
\downarrow \text{forget} & & \downarrow B A \rightarrow B B \\
\mathrm{Cat}^{\mathrm{LMod}_A(\mathcal{M})} & \xrightarrow{\quad} & \mathrm{LMod}_{B_A}(\mathrm{Cat}^{\mathcal{M}})
\end{array}$$

- By combining the above two commutative diagrams together, there is a commutative diagram:

$$\begin{array}{ccc}
\mathrm{Cat}^{\mathrm{LMod}_B(\mathcal{M})} & \xrightarrow{\quad} & \mathrm{LMod}_B(\mathrm{Cat}^{\mathcal{M}}) \\
\downarrow (F_{B/A})_* & & \downarrow B A \rightarrow B B \\
\mathrm{Cat}^{\mathrm{LMod}_A(\mathcal{M})} & \xrightarrow{\quad} & \mathrm{LMod}_{B_A}(\mathrm{Cat}^{\mathcal{M}})
\end{array}$$

where the left vertical morphism is induced by the transfer of enrichment along the forgetful functor $\mathrm{LMod}_B(\mathcal{M}) \rightarrow \mathrm{LMod}_A(\mathcal{M})$, and the right vertical morphism is induced by the forgetful functor $\mathrm{LMod}_B(\mathrm{Cat}^{\mathcal{M}}) \rightarrow \mathrm{LMod}_{B_A}(\mathrm{Cat}^{\mathcal{M}})$ along $B A \rightarrow B B$, and the horizontal morphisms are equivalences by [corollary 2.4.2](#).

2.5 $\mathrm{LMod}_{\mathrm{Span}_1(\mathcal{C})}(\mathrm{Cat})_{\mathrm{closed}} \hookrightarrow \mathrm{Cat}^{\mathrm{Span}_1(\mathcal{C})}$

This subsection is an expansion of the [\[37, section 2\]](#).

By [\[15, theorem 7.21\]](#), there is a lax symmetric monoidal functor

$$\mathrm{LMod}_{\mathrm{Span}_1(\mathcal{C})}(\mathrm{Cat}) \rightarrow \mathrm{Cat}^{\mathcal{P}(\mathrm{Span}_1(\mathcal{C}))}$$

which is a non-full subcategory inclusion.

Consider the full subcategory $\mathrm{LMod}_{\mathrm{Span}_1(\mathcal{C})}(\mathrm{Cat})_{\mathrm{closed}} \subset \mathrm{LMod}_{\mathrm{Span}_1(\mathcal{C})}(\mathrm{Cat})$ consisting of those left $\mathrm{Span}_1(\mathcal{C})$ -modules with closed action, then one can show that:

Lemma 2.5.1. *The embedding $\mathrm{LMod}_{\mathrm{Span}_1(\mathcal{C})}(\mathrm{Cat})_{\mathrm{closed}} \hookrightarrow \mathrm{LMod}_{\mathrm{Span}_1(\mathcal{C})}(\mathrm{Cat})$ is closed under tensor product $\otimes_{\mathrm{Span}_1(\mathcal{C})}$.*

Proof. It suffices to show that for $\mathcal{M}, \mathcal{N} \in \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}}$, the relative tensor product $\mathcal{M} \otimes_{\text{Span}_1(\mathcal{C})} \mathcal{N}$ is again a closed $\text{Span}_1(\mathcal{C})$ -module.

- Noticing that the canonical map $\mathcal{M} \rightarrow \mathcal{N} \rightarrow \mathcal{M} \otimes_{\text{Span}_1(\mathcal{C})} \mathcal{N}$ is surjective at the level of underlying spaces since it's the last term in the bar construction, thus it suffices to show that for any $m \in \mathcal{M}, n \in \mathcal{N}, m \boxtimes n \in \mathcal{M} \otimes_{\text{Span}_1(\mathcal{C})} \mathcal{N}$, the morphism $\text{Span}_1(\mathcal{C}) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}) \rightarrow \mathcal{M} \otimes_{\text{Span}_1(\mathcal{C})} \mathcal{N}$ induced by $\text{Span}_1(\mathcal{C}) \rightarrow \mathcal{M}$ specifying the object m and $\text{Span}_1(\mathcal{C}) \rightarrow \mathcal{N}$ specifying the object n , admits a right adjoint $\underline{\text{Hom}}_{\mathcal{M} \otimes_{\text{Span}_1(\mathcal{C})} \mathcal{N}}(m \boxtimes n, -)$.
- Noticing that the symmetric monoidal functor $\text{Cat} \rightarrow \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$ makes $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$ into a symmetric monoidal $(\infty, 2)$ category, thus the tensor product $\otimes_{\text{Span}_1(\mathcal{C})}$ preserves adjunction. And it remains to show that the right adjoints

$$\underline{\text{Hom}}_{\mathcal{M}}(m, -) : \mathcal{M} \rightarrow \text{Span}_1(\mathcal{C}), \underline{\text{Hom}}_{\mathcal{N}}(n, -) : \mathcal{N} \rightarrow \text{Span}_1(\mathcal{C})$$

are $\text{Span}_1(\mathcal{C})$ -linear functors, which is true by noting that every object in $\text{Span}_1(\mathcal{C})$ has a dual: for any $a \in \text{Span}_1(\mathcal{C})$,

$$\text{Hom}_{\mathcal{M}}(m, a \otimes (-)) \cong \text{Hom}_{\mathcal{M}}(m \otimes a^\vee, (-)) \cong \underline{\text{Hom}}_{\text{Span}_1(\mathcal{C})}(a^\vee \underline{\text{Hom}}_{\mathcal{M}}(m, (-)) \cong a \otimes \underline{\text{Hom}}_{\mathcal{M}}(m, (-))$$

□

Remark 2.5.2. Especially, by the above remark, we can see that

$$\underline{\text{Hom}}_{\mathcal{M} \otimes_{\text{Span}_1(\mathcal{C})} \mathcal{N}}(m \boxtimes n, x \boxtimes y) \cong \underline{\text{Hom}}_{\mathcal{M}}(m, x) \boxtimes \underline{\text{Hom}}_{\mathcal{N}}(n, y) \in \text{Span}_1(\mathcal{C}) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}) \cong \underline{\text{Hom}}_{\mathcal{M}}(m, x) \times \underline{\text{Hom}}_{\mathcal{N}}(n, y)$$

.

Thus, we can enhance $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}}$ as a symmetric monoidal full subcategory of $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$, and we get a lax symmetric monoidal functor

$$\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}} \rightarrow \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$$

.

Noticing that the symmetric monoidal functor $(\mathcal{L}_{\text{Span}_1(\mathcal{C})})_* : \text{Cat}^{\text{Span}_1(\mathcal{C})} \hookrightarrow \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$ is fully faithful by [16, corollary 4.74], thus the above inclusion factors as

$$\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}} \rightarrow \text{Cat}^{\text{Span}_1(\mathcal{C})}$$

.

Moreover, one can show that:

Lemma 2.5.3. *The inclusion $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}} \rightarrow \text{Cat}^{\text{Span}_1(\mathcal{C})}$ is a fully faithful embedding.*

Proof. • We work in the model of enriched category by weakly left tensored categories, and it suffices to show that any lax $\text{Span}_1(\mathcal{C})$ left module morphism between $\mathcal{M}, \mathcal{N} \in \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}}$ is also a strict left module morphism: that is, an operad morphism $F : (\text{Span}_1(\mathcal{C}), \mathcal{M}) \rightarrow (\text{Span}_1(\mathcal{C}), \mathcal{N})$ over $\text{LM}^\otimes$ which is identity over $\text{Ass}^\otimes \hookrightarrow \text{LM}^\otimes$ will automatically preserve the coCartesian liftings over the active morphism in $\text{LM}^\otimes$.

- It suffices to show that for any $a \in \text{Span}_1(\mathcal{C})$, $m \in \mathcal{M}$, the induced morphism

$$\rho : a \otimes F(m) \rightarrow F(a \otimes m)$$

is an equivalence.

- We construct the inverse map of ρ $\tau : F(a \otimes m) \rightarrow a \otimes F(m)$ by the dual map $a^\vee \otimes F(a \otimes m) \rightarrow F(a^\vee \otimes a \otimes m) \xrightarrow{F(\text{coev}) \otimes \text{id}} F(m)$, and it remains to show that the following compositions are equivalent to identity maps

$$\begin{aligned} a \otimes F(m) &\xrightarrow{\rho} F(a \otimes m) \xrightarrow{\tau} a \otimes F(m) \\ F(a \otimes m) &\xrightarrow{\tau} a \otimes F(m) \xrightarrow{\rho} F(a \otimes m) \end{aligned}$$

.

- For the first composition of morphisms, notice the following commutative diagrams:

$$\begin{array}{ccc} a \otimes F(x) & \xrightarrow{\text{coev}} & (a \otimes a^\vee) \otimes a \otimes F(x) \\ \downarrow \rho & & \downarrow \\ F(a \otimes x) & \longrightarrow & (a \otimes a^\vee) \otimes F(a \otimes x) \end{array}$$

and

$$\begin{array}{ccc} (a \otimes a^\vee) \otimes a \otimes F(x) & \longrightarrow & a \otimes 1 \otimes F(x) \\ \downarrow \cong & \nearrow \text{ev} & \downarrow \cong \\ a \otimes (a^\vee \otimes a) \otimes F(x) & & a \otimes F(1 \otimes x) \\ \downarrow & \searrow \text{ev} & \\ a \otimes F(a^\vee \otimes a \otimes x) & \longrightarrow & a \otimes F(1 \otimes x) \end{array}$$

, which combine into

$$\begin{array}{ccccccc} a \otimes F(x) & \xrightarrow{\text{coev}} & (a \otimes a^\vee) \otimes a \otimes F(x) & & & & \\ \downarrow \rho & & \downarrow & & \searrow \text{ev} & & \\ F(a \otimes x) & \longrightarrow & (a \otimes a^\vee) \otimes F(a \otimes x) & \longrightarrow & a \otimes F(a^\vee \otimes a \otimes x) & \longrightarrow & a \otimes F(x) \\ & & & \nearrow \tau & & & \end{array}$$

- For the second composition, consider the following commutative diagrams:

$$\begin{array}{ccc} 1 \otimes F(a \otimes x) & \xrightarrow{\cong} & F(1 \otimes a \otimes x) \\ \text{coev} \downarrow & & \text{coev} \downarrow \\ (a \otimes a^\vee) \otimes F(a \otimes x) & \longrightarrow & F((a \otimes a^\vee) \otimes a \otimes x) \end{array}$$

and

$$\begin{array}{ccc} a \otimes F(a^\vee \otimes a \otimes x) & \xrightarrow{\text{ev}} & a \otimes F(x) \\ \downarrow & & \downarrow \rho \\ F(a \otimes (a^\vee \otimes a) \otimes x) & \xrightarrow{\text{ev}} & F(a \otimes x) \end{array}$$

which combine into

$$\begin{array}{ccccccc}
 & & a \otimes F(a^\vee \otimes a \otimes x) & \xrightarrow{coev} & a \otimes F(1 \otimes x) & & \\
 & \nearrow & \downarrow & & \searrow \rho & & \\
 (a \otimes a^\vee) \otimes F(a \otimes x) & \xrightarrow{\quad} & F((a \otimes a^\vee) \otimes a \otimes x) & \xrightarrow{\cong} & F(a \otimes (a^\vee \otimes a) \otimes x) & \xrightarrow{coev} & F(a \otimes x) \\
 \uparrow ev & & \nearrow ev & & & & \\
 1 \otimes F(a \otimes x) & \cong & F(1 \otimes a \otimes x) & & & &
 \end{array}$$

where the compositions of the morphisms from $1 \otimes F(a \otimes x)$ at the bottom to $F(a \otimes x)$ at the right hand side along the upper path give us $\rho \circ \tau$, and the compositions along the lower path give us id .

□

2.6 $\text{Span}_n^{enr}(\mathcal{C}) \in n \text{Cat}^{\text{Span}_1(\mathcal{C})}$

The following construction is [37, section 2]:

Notation 2.6.1. Given a symmetric monoidal category $\mathcal{V}$, $\text{Cat}^\mathcal{V}$ is again symmetric monoidal,

- (1) We set $k \text{Cat}^\mathcal{V}$ as $\mathcal{V}$ when $k = 0$, and $(k + 1) \text{Cat}^\mathcal{V}$ as $\text{Cat}^k \text{Cat}^\mathcal{V}$ for $k \geq 0$.
- (2) We set $B^0 = \text{id}$ and $B^{k+1}(-) = B \circ B^k(-)$ for the symmetric monoidal adjunction $B : \text{Alg}(\mathcal{V}) \rightleftarrows (\text{Cat}^\mathcal{V})_{*/} : \Omega$.

Construction 2.6.2. Now consider $B \Omega \text{Span}_2(\mathcal{C}) \cong B \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_2(\mathcal{C}) \in \text{CAlg}(\text{Cat}_2)$ induced by the symmetric monoidal adjunction $B \dashv \Omega$, by [proposition 2.3.5](#), $B \text{Span}_1(\mathcal{C})$ here is equipped with the commutative algebra structure induced by $\text{Span}_1(\mathcal{C}) \in \text{CAlg}(\text{Cat})$ under the symmetric monoidal functor B .

Under the equivalence $\text{CAlg}(\text{LMod}_{B \text{Span}_1(\mathcal{C})}(\text{Cat}_2)) \cong \text{CAlg}(\text{Cat}_2)_{B \text{Span}_1(\mathcal{C})/}$, the map $B \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_2(\mathcal{C})$ makes $\text{Span}_2(\mathcal{C})$ into a commutative algebra object in $\text{LMod}_{B \text{Span}_1(\mathcal{C})}(\text{Cat}_2) \cong \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}$, denoted as $\text{Span}_2^{enr}(\mathcal{C}) \in \text{CAlg}(\text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})})$.

Inductively, for $n \geq 2$, consider $B^{n-1} \text{Span}_1(\mathcal{C}) \hookrightarrow \text{Span}_n(\mathcal{C})$. This makes $\text{Span}_n(\mathcal{C})$ into a commutative algebra object in $\text{LMod}_{B^{n-1} \text{Span}_1(\mathcal{C})}(\text{Cat}_n) \cong (n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}$, denoted as $\text{Span}_n^{enr}(\mathcal{C}) \in \text{CAlg}((n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})})$.

Remark 2.6.3. If we further consider the symmetric monoidal (∞, n) functors:

$$* = B^{n-1} * \rightarrow B^{n-1} \text{Span}_1(\mathcal{C}) \hookrightarrow \text{Span}_n(\mathcal{C})$$

, by [remark 2.4.3](#), the equivalence $\text{CAlg}(\text{Cat}^{\text{LMod}_A(\mathcal{M})}) \cong \text{CAlg}(\text{LMod}_{B A}(\text{Cat}^\mathcal{M})) \cong \text{CAlg}(\text{Cat}^\mathcal{M})_{B A/}$ is functorial, and we get the following commutative diagram:

$$\begin{array}{ccc}
 \text{Span}_2^{enr}(\mathcal{C}) \in \text{CAlg}(\text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}) & \xrightarrow{\cong} & \text{CAlg}(\text{Cat}_2)_{B \text{Span}_1(\mathcal{C})/} \ni (B \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_2(\mathcal{C})) \\
 \downarrow & & \downarrow * \rightarrow B \text{Span}_1(\mathcal{C}) \\
 \text{CAlg}(\text{Cat}^{\text{Cat}} = \text{Cat}_2) & \xrightarrow{\cong} & \text{CAlg}(\text{Cat}_2)_{*/} \ni \text{Span}_2(\mathcal{C})
 \end{array}$$

especially, this implies that under the transfer of enrichment along the forgetful functor $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}) \rightarrow \text{Cat}$, we recover the symmetric monoidal $(\infty, 2)$ category $\text{Span}_2(\mathcal{C}) \in \text{CAlg}(\text{Cat}_2)$. Inductively, one can show that

$$\begin{array}{ccc} \text{Span}_n^{\text{enr}}(\mathcal{C}) \in \text{CAlg}((n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}) & \xrightarrow{\cong} & \text{CAlg}(\text{Cat}_n)_{B^{n-1} \text{Span}_1(\mathcal{C})/} \ni (B^{n-1} \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_n(\mathcal{C})) \\ \downarrow & & \downarrow \scriptstyle * \rightarrow B \text{Span}_1(\mathcal{C}) \\ \text{CAlg}(\text{Cat}^{\text{Cat}_{n-1}} = \text{Cat}_n) & \xrightarrow{\cong} & \text{CAlg}(\text{Cat}_n)_{*/} \ni \text{Span}_n(\mathcal{C}) \end{array}$$

, thus under the transfer of enrichment along the forgetful functor $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}) \rightarrow \text{Cat}$, we can recover the symmetric monoidal (∞, n) category structure on $\text{Span}_n(\mathcal{C})$ from $\text{Span}_n^{\text{enr}}(\mathcal{C}) \in \text{CAlg}((n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})})$.

Now consider the fully faithful symmetric monoidal embedding $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}} \hookrightarrow \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$. We can check that

Lemma 2.6.4. $\text{Span}_n^{\text{enr}}(\mathcal{C}) \in (n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}$ further lands inside the full subcategory $\text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}}} \hookrightarrow \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}$.

Before the proof, one need to make the following observations about how we recover the left A -module structure on the enriched Hom $\mathcal{C}(x, y)$ for a given $B A \rightarrow \mathcal{C} \in \text{CAlg}(\text{Cat}^{\mathcal{M}})$ under the equivalence $\text{CAlg}(\text{Cat}^{\text{LMod}_A(\mathcal{M})}) \cong \text{CAlg}(\text{Cat}^{\mathcal{M}})_{B A/}$ in [corollary 2.4.2](#).

Observation 2.6.5. • Given $\mathcal{C} \in \text{CAlg}(\text{Cat}^{\text{LMod}_A(\mathcal{M})})$, there is a multiplication map for $\mathcal{C}$ in $\text{Cat}^{\text{LMod}_A(\mathcal{M})}$:

$$m_{\mathcal{C}}^A : \mathcal{C} \otimes_A \mathcal{C} \rightarrow \mathcal{C}$$

where $\mathcal{C} \otimes_A \mathcal{C}$ is the tensor product for the symmetric monoidal category $\text{Cat}^{\text{LMod}_A(\mathcal{M})}$, for objects $x, y \in \mathcal{C}$, there is an induced morphism of enriched Homs in $\text{LMod}_A(\mathcal{M})$:

$$\text{act}_{x,y}^A : (\mathcal{C} \otimes_A \mathcal{C})(1 \otimes x, 1 \otimes y) \cong \mathcal{C}(1, 1) \otimes_A \mathcal{C}(x, y) \rightarrow \mathcal{C}(x, y)$$

where 1 is the unit object in $\mathcal{C}$.

- Under the lax symmetric monoidal functor $\text{Cat}^{\text{LMod}_A(\mathcal{M})} \rightarrow \text{Cat}^{\mathcal{M}}$, the image of $\mathcal{C}$ in $\text{Cat}^{\mathcal{M}}$ is again a commutative algebra object, and we denote its multiplication map in $\text{Cat}^{\mathcal{M}}$ as $m_{\mathcal{C}}$. There is a factorization for the map $m_{\mathcal{C}}$:

$$m_{\mathcal{C}} : \mathcal{C} \otimes \mathcal{C} \rightarrow \mathcal{C} \otimes_A \mathcal{C} \xrightarrow{m_{\mathcal{C}}^A} \mathcal{C}$$

, and there is an induced morphism of enriched Homs in $\mathcal{M}$:

$$(!) \quad \text{act}_{x,y} : \mathcal{C}(1, 1) \otimes \mathcal{C}(x, y) \xrightarrow{s} \mathcal{C}(1, 1) \otimes_A \mathcal{C}(x, y) \xrightarrow{\text{act}_{x,y}^A} \mathcal{C}(x, y)$$

(without abuse of notation, we again denote the image of $\mathcal{C}(1, 1), \mathcal{C}(x, y) \in \text{LMod}_A(\mathcal{M})$ in $\mathcal{M}$ as $\mathcal{C}(1, 1)$ and $\mathcal{C}(x, y)$) where $\mathcal{C}(1, 1) \otimes \mathcal{C}(x, y)$ is the tensor product inside the symmetric monoidal category $\mathcal{M}$, and $\mathcal{C}(1, 1) \otimes_A \mathcal{C}(x, y)$ is the tensor product inside the symmetric monoidal category $\text{LMod}_A(\text{Cat})$.

- Moreover, the map s in (!) is the canonical factorization induced by the lax symmetric monoidal structure on the functor $\mathbf{LMod}_A(\mathcal{M}) \rightarrow \mathcal{M}$. To show this, we work in the model of enriched categories by categorical algebras:

(1) Noticing that for the symmetric monoidal adjunction

$$loc_{\mathcal{M}} : \mathbf{Alg}_{cat}(\mathcal{M}) \rightleftarrows \mathbf{Cat}^{\mathcal{M}} : i_{\mathcal{M}}$$

and

$$loc_{\mathbf{LMod}_A(\mathcal{M})} : \mathbf{Alg}_{cat}(\mathbf{LMod}_A(\mathcal{M})) \rightleftarrows \mathbf{Cat}^{\mathbf{LMod}_A(\mathcal{M})} : i_{\mathbf{LMod}_A(\mathcal{M})}$$

, the right adjoints are usually only lax symmetric monoidal, that is,

$$(!) \quad \mathcal{C} \otimes^{cat} \mathcal{C} \rightarrow \mathcal{C} \otimes \mathcal{C}, \mathcal{C} \otimes_A^{cat} \mathcal{C} \rightarrow \mathcal{C} \otimes_A \mathcal{C}$$

is not an equivalence in the category of categorical algebras (where $\otimes^{cat}$ denotes the tensor product in the symmetric monoidal category $\mathbf{Alg}_{cat}(\mathcal{M})$ and $\otimes_A^{cat}$ denotes the tensor product of symmetric monoidal category $\mathbf{Alg}_{cat}(\mathbf{LMod}_A(\mathcal{M}))$).

- (2) However, the above induced morphisms ((!)) is a local equivalence for $loc_{\mathcal{M}} \dashv i_{\mathcal{M}}$ by [10, proposition 5.7.14], [10, proposition 5.7.17] and [10, lemma 7.5], that is, they are enriched fully faithful and essential surjective. Consider the following diagram in $\mathbf{Alg}_{cat}(\mathcal{M})$:

$$\begin{array}{ccccc} \mathcal{C} \otimes^{cat} \mathcal{C} & \longrightarrow & \mathcal{C} \otimes_A^{cat} \mathcal{C} & \xrightarrow{m_e^{A,cat}} & \mathcal{C} \\ \downarrow & & \downarrow & & \downarrow \cong \\ \mathcal{C} \otimes \mathcal{C} & \longrightarrow & \mathcal{C} \otimes_A \mathcal{C} & \xrightarrow{m_e^A} & \mathcal{C} \\ & \searrow m_e & & & \end{array}$$

where all the vertical arrows are the localizations and the upper horizontal arrows are the factorization of the multiplication map for $\mathcal{C}$ regarded as a commutative algebra object in $\mathbf{Alg}_{cat}(\mathbf{LMod}_A(\mathcal{M})) \rightarrow \mathbf{Alg}_{cat}(\mathcal{M})$, and the vertical arrows induce equivalences for the enriched Homs.

Thus we may assume that the morphism (!) is induced by the factorization for the multiplication map m_e^{cat} of $\mathcal{C}$ under the lax symmetric monoidal functor $\mathbf{Alg}_{cat}(\mathbf{LMod}_A(\mathcal{M})) \rightarrow \mathbf{Alg}_{cat}(\mathcal{M})$:

$$m_e^{cat} : \mathcal{C} \otimes^{cat} \mathcal{C} \rightarrow \mathcal{C} \otimes_A^{cat} \mathcal{C} \xrightarrow{m_e^{A,cat}} \mathcal{C}$$

- (3) Noticing that this factorization map is induced by the Beck-Chevalley map by taking the vertical right adjoints of the following commutative diagram:

$$\begin{array}{ccccc} \mathbf{Alg}_{cat}(\mathcal{M}) \times \mathbf{Alg}_{cat}(\mathcal{M}) & \xrightarrow{\boxtimes_{\mathcal{M}}} & \mathbf{Alg}_{cat}(\mathcal{M} \times \mathcal{M}) & \xrightarrow{(\text{mult}_{\mathcal{M}})_*} & \mathbf{Alg}_{cat}(\mathcal{M}) \\ ((-)\otimes A)_* \times ((-)\otimes A)_* \downarrow & & (((-)\otimes A) \times ((-)\otimes A))_* \downarrow & & \downarrow ((-)\otimes A)_* \\ \mathbf{Alg}_{cat}(\mathbf{LMod}_A(\mathcal{M})) \times \mathbf{Alg}_{cat}(\mathbf{LMod}_A(\mathcal{M})) & \xrightarrow{\boxtimes_{\mathbf{LMod}_A(\mathcal{M})}} & \mathbf{Alg}_{cat}(\mathbf{LMod}_A(\mathcal{M}) \times \mathbf{LMod}_A(\mathcal{M})) & \xrightarrow{(\text{mult}_{\mathbf{LMod}_A(\mathcal{M})})_*} & \mathbf{Alg}_{cat}(\mathbf{LMod}_A(\mathcal{M})) \end{array}$$

, where $\boxtimes_{\mathcal{M}}$ sends a pair of categorical algebras $F : \mathbf{Ass}_X \rightarrow \mathcal{M}, G : \mathbf{Ass}_Y \rightarrow \mathcal{M}$ to $F \times_{\mathbf{Ass}} G : \mathbf{Ass}_{X \times Y} \rightarrow \mathcal{M} \times \mathcal{M}$, so is $\boxtimes_{\mathbf{LMod}_A(\mathcal{M})}$.

- (4) For $(\mathcal{C}, \mathcal{C}) \in \text{Alg}_{\text{cat}}(\text{LMod}_A(\mathcal{M})) \times \text{Alg}_{\text{cat}}(\text{LMod}_A(\mathcal{M}))$, let $F : \text{Ass}_X \rightarrow \text{LMod}_A(\mathcal{M})$ denote the categorical algebra $\mathcal{C}$, and $(1, x), (1, y) \in X \times X$ are objects of $(\mathcal{C}, \mathcal{C})$ (1 is the unit object), then the Beck-Chevalley map evaluated at $(\mathcal{C}, \mathcal{C})$ agrees with the Beck-Chevalley map associated to the diagram of fibers:

$$\begin{array}{ccccc}
 \text{Alg}_{\text{Ass}_X}(\mathcal{M}) \times \text{Alg}_{\text{Ass}_X}(\mathcal{M}) & \xrightarrow{\boxtimes_{\mathcal{M}}} & \text{Alg}_{\text{Ass}_{X \times X}}(\mathcal{M} \times \mathcal{M}) & \xrightarrow{(\text{mult}_{\mathcal{M}})_*} & \text{Alg}_{\text{Ass}_{X \times X}}(\mathcal{M}) \\
 ((-)\otimes A)_* \times ((-)\otimes A)_* \downarrow & & (((-)\otimes A) \times ((-)\otimes A))_* \downarrow & & \downarrow ((-)\otimes A)_* \\
 \text{Alg}_{\text{Ass}_{X \times X}}(\text{LMod}_A(\mathcal{M})) \times \text{Alg}_{\text{Ass}_X}(\text{LMod}_A(\mathcal{M})) & \xrightarrow{\boxtimes_{\text{LMod}_A(\mathcal{M})}} & \text{Alg}_{\text{Ass}_{X \times X}}(\text{LMod}_A(\mathcal{M}) \times \text{LMod}_A(\mathcal{M})) & \xrightarrow{(\text{mult}_{\text{LMod}_A(\mathcal{M})})_*} & \text{Alg}_{\text{Ass}_{X \times X}}(\text{LMod}_A(\mathcal{M}))
 \end{array}$$

- (5) By [10, section 3.4], there is a trivial generalized non-symmetric operad $\text{Ass}_X^{\text{triv}} \rightarrow \text{Ass}_X$ associated to Ass_X satisfying the property that $\text{Alg}_{\text{Ass}_X^{\text{triv}}}(\mathcal{V}) \cong \text{Fun}(X \times X, \mathcal{V})$ for any non-symmetric operad $\mathcal{V}$. Thus by taking the induced morphism $\text{Ass}_X^{\text{triv}} \rightarrow \text{Ass}_X$, the above diagram translates into

$$\begin{array}{ccccc}
 \text{Fun}(X^{\times 2}, \mathcal{M}) \times \text{Fun}(X^{\times 2}, \mathcal{M}) & \xrightarrow{\boxtimes_{\mathcal{M}}} & \text{Fun}(X^{\times 4}, \mathcal{M} \times \mathcal{M}) & \xrightarrow{(\text{mult}_{\mathcal{M}})_*} & \text{Fun}(X^{\times 4}, \mathcal{M}) \\
 ((-)\otimes A)_* \times ((-)\otimes A)_* \downarrow & & (((-)\otimes A) \times ((-)\otimes A))_* \downarrow & & \downarrow ((-)\otimes A)_* \\
 \text{Fun}(X^{\times 2}, \text{LMod}_A(\mathcal{M})) \times \text{Fun}(X^{\times 2}, \text{LMod}_A(\mathcal{M})) & \xrightarrow{\boxtimes_{\text{LMod}_A(\mathcal{M})}} & \text{Fun}(X^{\times 4}, \text{LMod}_A(\mathcal{M}) \times \text{LMod}_A(\mathcal{M})) & \xrightarrow{(\text{mult}_{\text{LMod}_A(\mathcal{M})})_*} & \text{Fun}(X^{\times 4}, \text{LMod}_A(\mathcal{M}))
 \end{array}$$

- (6) The map s between enriched Homs in (!) is the map by evaluating the Beck-Chevalley map associated to the above diagram at $(1, 1) \in X \times X, (x, y) \in X \times X$, which boils down to the Beck-Chevalley map (by taking vertical right adjoints) of the diagram:

$$\begin{array}{ccc}
 \mathcal{M} \times \mathcal{M} & \xrightarrow{\text{mult}_{\mathcal{M}}} & \mathcal{M} \\
 ((-)\otimes A) \times ((-)\otimes A) \downarrow & & (-)\otimes A \downarrow \\
 \text{LMod}_A(\mathcal{M}) \times \text{LMod}_A(\mathcal{M}) & \xrightarrow{\text{mult}_{\text{LMod}_A(\mathcal{M})}} & \text{LMod}_A(\mathcal{M})
 \end{array}$$

, which value at $(x, y) \in \text{LMod}_A(\mathcal{M}) \times \text{LMod}_A(\mathcal{M})$ is precisely the canonical map $x \times y \rightarrow x \otimes_A y$ for $x, y \in \text{LMod}_A(\mathcal{M})$, induced by the lax symmetric monoidal structure functor $\text{LMod}_A(\mathcal{M}) \rightarrow \mathcal{M}$.

- Consider the structure morphism $BA \rightarrow \mathcal{C} \in (\text{CAlg}(\text{Cat}^{\text{LMod}_A(\mathcal{M})}) \rightarrow \text{CAlg}(\text{Cat}^{\mathcal{M}}))$, and the commutative diagram induced by the multiplication maps:

$$\begin{array}{ccccc}
 BA \otimes \mathcal{C} & \longrightarrow & BA \otimes_A \mathcal{C} & & \\
 \downarrow & & \downarrow & \searrow \cong & \\
 \mathcal{C} \otimes \mathcal{C} & \longrightarrow & \mathcal{C} \otimes_A \mathcal{C} & \xrightarrow{m_{\mathcal{C}}^A} & \mathcal{C} \\
 & \searrow m_{\mathcal{C}} & & \nearrow &
 \end{array}$$

where the upper horizontal map is the BA -module structure map on $\mathcal{C}$ under the equivalence $\text{Cat}^{\text{LMod}_A(\mathcal{M})} \cong \text{LMod}_{BA}(\text{Cat}^{\mathcal{M}})$

- For any $x, y \in \mathcal{C}$, the above diagram gives rise to a the following commutative diagram:

$$\begin{array}{ccccc}
A \otimes \mathcal{C}(x, y) & \xrightarrow{s_A} & A \otimes_A \mathcal{C}(x, y) & & \\
\downarrow t \otimes \text{id} & & \downarrow t \otimes_A \text{id} & \searrow \cong & \\
\mathcal{C}(1, 1) \otimes \mathcal{C}(x, y) & \xrightarrow{s} & \mathcal{C}(1, 1) \otimes_A \mathcal{C}(x, y) & \xrightarrow{\text{act}_{x,y}^A} & \mathcal{C}(x, y)
\end{array}$$

where $t : A \rightarrow \mathcal{C}(1, 1)$ is induced by the structure map $B A \rightarrow \mathcal{C} \in \text{CAlg}(\text{Cat}^{\text{LMod}_A(\mathcal{M})})$, and s_A is the factorization map induced by the lax symmetric monoidal structure on $\text{LMod}_A(\mathcal{M}) \rightarrow \mathcal{M}$.

- Especially, s_A recovers the left A action map for $\mathcal{C}(x, y) \in \text{LMod}_A(\mathcal{M})$. In conclusion, given a morphism of commutative algebra objects $B A \rightarrow \mathcal{C} \in \text{CAlg}(\text{Cat}^{\mathcal{M}})$, we can recover the left A module structure on $\mathcal{C}(x, y)$ by the composition of $A \rightarrow \mathcal{C}(1, 1)$ with the map $\mathcal{C}(1, 1) \otimes \mathcal{C}(x, y) \cong (\mathcal{C} \otimes \mathcal{C})(1 \otimes x, 1 \otimes y) \rightarrow \mathcal{C}(x, y)$ induced by the multiplication map $m_{\mathcal{C}} : \mathcal{C} \otimes \mathcal{C} \rightarrow \mathcal{C}$.

Proof of Lemma 2.6.4. We show this by induction: for $n = 2$,

- The action map for $\text{Span}_2^{\text{enr}}(\mathcal{C}) \in \text{LMod}_{B \text{Span}_1(\mathcal{C})}(\text{Cat}_2)$ is induced by

$$B \text{Span}_1(\mathcal{C}) \times \text{Span}_2(\mathcal{C}) \hookrightarrow \text{Span}_2(\mathcal{C}) \times \text{Span}_2(\mathcal{C}) \xrightarrow{\text{mult}} \text{Span}_2(\mathcal{C})$$

where the first arrow is the full $(\infty, 2)$ category embedding, and $\text{mult} : \text{Span}_2(\mathcal{C}) \times \text{Span}_2(\mathcal{C}) \cong \text{Span}_2(\mathcal{C} \times \mathcal{C}) \rightarrow \text{Span}_2(\mathcal{C})$ is induced by $\times : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$.

- Thus for $x, y \in \text{Span}_2(\mathcal{C})$, under the equivalences

$$\underline{\text{Hom}}_{\text{Span}_2(\mathcal{C} \times \mathcal{C})}((*, x), (*, y)) \cong \text{Span}_1((\mathcal{C} \times \mathcal{C})_{/(*, x \times y)}) \cong \text{Span}_1(\mathcal{C}) \times \text{Span}_1(\mathcal{C}_{/x \times y})$$

and

$$\underline{\text{Hom}}_{\text{Span}_2(\mathcal{C})}(* \times x, * \times y) \cong \text{Span}_1(\mathcal{C}_{x \times y})$$

, the action map

$$\text{act} : \text{Span}_1(\mathcal{C}) \times \text{Span}_1(\mathcal{C}_{/x \times y}) \rightarrow \text{Span}_1(\mathcal{C}_{/x \times y})$$

is induced by $\times : (\mathcal{C} \times \mathcal{C})_{/(*, x \times y)} \rightarrow \mathcal{C}_{/x \times y}$.

- Thus the action is closed: for any $t \in \text{Span}_1(\mathcal{C}_{/x \times y})$, $\text{act}(-, t) : \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_1(\mathcal{C}_{/x \times y})$ is induced by $(-) \times t : \mathcal{C} \rightarrow \mathcal{C}_{/x \times y}$, which admits a right adjoint by

$$\text{Span}_1(\mathcal{C}_{/x \times y}) \xrightarrow{(-) \times_{x \times y} t} \text{Span}_1(\mathcal{C}_{/t}) \xrightarrow{\text{forget}} \text{Span}_1(\mathcal{C}).$$

Assume that for $k = n \geq 2$, the morphism $B^{n-1} \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_n(\mathcal{C}) \in \text{CAlg}(\text{Cat}_n)$ makes $\text{Span}_n^{\text{enr}}(\mathcal{C}) \in (n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}$ lands in $(n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})^{\text{closed}}}$, then for $k = n+1$:

- For any $x, y \in \mathcal{C}$, the left module structure map gives us:

$$B^n \text{Span}_1(\mathcal{C}) \times \text{Span}_{n+1}(\mathcal{C}) \rightarrow \text{Span}_{n+1}(\mathcal{C}) \times \text{Span}_{n+1}(\mathcal{C}) \cong \text{Span}_{n+1}(\mathcal{C} \times \mathcal{C}) \xrightarrow{\text{mult}} \text{Span}_{n+1}(\mathcal{C})$$

- Thus the induced action of enriched Homs is given by:

$$B^{n-1} \text{Span}_1(\mathcal{C}) \times \text{Span}_n(\mathcal{C}_{/x \times y}) \rightarrow \text{Span}_n(\mathcal{C}) \times \text{Span}_n(\mathcal{C}_{/x \times y}) \rightarrow \text{Span}_n(\mathcal{C}_{/x \times y})$$

, similar to the case when $n = 2$, the last action map is induced by the symmetric monoidal (∞, n) functor $(-) \times (x \times y) : \text{Span}_n(\mathcal{C}) \rightarrow \text{Span}_n(\mathcal{C}_{/x \times y})$, thus the $B^{n-1} \text{Span}_1(\mathcal{C})$ action is induced by :

$$B^{n-1} \text{Span}_1(\mathcal{C}) \rightarrow B^{n-1} \text{Span}_1(\mathcal{C}_{/x \times y}) \hookrightarrow \text{Span}_n(\mathcal{C}_{/x \times y})$$

where the first arrow is $(-) \times (x \times y) : \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_1(\mathcal{C}_{/x \times y})$.

- By assumption, the left $B^{n-1} \text{Span}_1(\mathcal{C}_{/x \times y})$ on $\text{Span}_n(\mathcal{C}_{/x \times y})$ makes $\text{Span}_n(\mathcal{C}_{/x \times y})$ enhance into an object $\text{Span}_n^{enr}(\mathcal{C}_{/x \times y}) \in (n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C}_{/x \times y})}(\text{Cat})^{\text{closed}}} \hookrightarrow (n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C}_{/x \times y})}(\text{Cat})}$, by the functoriality in [remark 2.4.3](#), there is a commutative diagram:

(□)

$$\begin{array}{ccc} \text{CAlg}((n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C}_{/x \times y})}(\text{Cat})}) & \xrightarrow{\quad} & \text{CAlg}(\text{Cat}_n)_{B^{n-1} \text{Span}_1(\mathcal{C}_{/x \times y})/} \\ \downarrow & & \downarrow \\ \text{CAlg}((n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}) & \xrightarrow{\quad} & \text{CAlg}(\text{Cat}_n)_{B^{n-1} \text{Span}_1(\mathcal{C})/} \end{array}$$

where the vertical maps are induced by the forgetful functor along $(-) \times (x \times y) : \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_1(\mathcal{C}_{/x \times y})$.

Thus it remains to show that the composition map

$$\text{LMod}_{\text{Span}_1(\mathcal{C}_{/x \times y})}(\text{Cat})^{\text{closed}} \hookrightarrow \text{LMod}_{\text{Span}_1(\mathcal{C}_{/x \times y})}(\text{Cat}) \xrightarrow{(-) \times (x \times y) : \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_1(\mathcal{C}_{/x \times y})} \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$$

factors through the full subcategory $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})^{\text{closed}} \hookrightarrow \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$, which is true since the functor $(-) \times (x \times y) : \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_1(\mathcal{C}_{/x \times y})$ admits a right adjoint functor induced by the slice forgetful functor $\mathcal{C}_{/x \times y} \rightarrow \mathcal{C}$. $\square$

By [\[15, theorem 7.21\]](#), there is a lax symmetric monoidal functor

$$\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}) \rightarrow \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$$

which is a non-full subcategory inclusion.

Proposition 2.6.6. *There exists a symmetric monoidal adjunction*

$$\text{loc} : \text{Cat}^{\text{Env}(\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}))} \rightleftarrows \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})} : i$$

induced by the symmetric monoidal adjunction $\text{loc} : \text{Env}(\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})) \rightleftarrows \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$:
i by [25, 2.2.4], there thus exists a lax symmetric monoidal functor

$$\text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})} \rightarrow \text{Cat}^{\text{Env}(\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}))} \rightarrow \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$$

. And by induction, there is a lax symmetric monoidal functor from $(n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})} \rightarrow n \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$.

Now consider the symmetric monoidal full subcategory embedding $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}} \hookrightarrow \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$, we have

Corollary 2.6.7. For $n \geq 2$ and $\mathcal{C} \in \text{Cat}^{pb}$, $\text{Span}_n^{enr}(\mathcal{C}) \in \text{CAlg}((n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})})$ lands inside $\text{CAlg}((n-1) \text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}}})$, thus lands inside $\text{CAlg}(n \text{Cat}^{\text{Span}_1(\mathcal{C})})$.

3 $\text{Span}_n^{enr}(\mathcal{C}) \rightarrow (n-1) \text{Cat}^{\text{Span}_1(\mathcal{C})}$

In this section, we assume that $n \geq 2$, and we show that there exists a lax symmetric monoidal (∞, n) functor $\text{Span}_{n-1}^{enr}(\mathcal{C}_{/(-)}) : \text{Span}_n^{enr}(\mathcal{C}) \rightarrow (n-1) \text{Cat}^{\text{Span}_1(\mathcal{C})}$.

3.1 $\Gamma : \mathcal{P}_{(n-1) \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}}(\text{Span}_n^{enr}(\mathcal{C})) \rightarrow (n-1) \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$

Assume $n \geq 2$ in this subsection.

By corollary 2.6.7, we have

$$\text{Span}_n^{enr}(\mathcal{C}) \in \text{CAlg}(n \text{Cat}^{\text{Span}_1(\mathcal{C})}) \hookrightarrow \text{CAlg}(n \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))})$$

Remark 3.1.1. • For a presentably monoidal category $\mathcal{V}$, $\text{Cat}^{\mathcal{V}}$ is again presentable by [16, proposition 2.60, corollary 4.62].

- Thus for any $n \geq 1$, $n \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$ is presentably symmetric monoidal category.

Notation 3.1.2. Let $\mathcal{V}$ denote the presentably symmetric monoidal category $(n-1) \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$.

Consider the enriched Yoneda embedding:

$$\mathbf{y}_{\mathcal{V}} : \text{Span}_n^{enr}(\mathcal{C}) \hookrightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{enr}(\mathcal{C}))$$

which lands in $\text{CAlg}(\text{Cat}^{\mathcal{V}})$ by [17, corollary 1.7] (The functor $\mathcal{P}_{\mathcal{V}}$ is a symmetric monoidal left adjoint of the inclusion $\text{LMod}_{\mathcal{V}}(\text{Pr}^L) \hookrightarrow \widehat{\text{Cat}}^{\mathcal{V}}$, and the unit map of the above symmetric monoidal adjunction is the enriched Yoneda embedding.)

Now consider $\mathcal{P}_{\mathcal{V}}(\text{Span}_n^{enr}(\mathcal{C})) \in \text{CAlg}(\text{LMod}_{\mathcal{V}}(\text{Pr}^L))$, consider the unit map

$$1 : \mathcal{V} \rightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{enr}(\mathcal{C}))$$

, then we have

Lemma 3.1.3. *The above symmetric monoidal functor*

$$1 : \mathcal{V} \rightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))$$

is an internal left adjoint in $\text{LMod}_{\mathcal{V}}(\text{Pr}^L)$.

Proof. • First noticing that the functor 1 can be equivalently regarded as the functor

$$\mathcal{P}_{\mathcal{V}}(\text{B } 1_{\mathcal{V}}) \rightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))$$

induced by the symmetric monoidal $\mathcal{V}$ -enriched functor

$$\text{B } 1_{\mathcal{V}} \cong 1_{\text{Cat}^{\mathcal{V}}} \rightarrow \text{Span}_n^{\text{enr}}(\mathcal{C})$$

in $\text{CAlg}(n \text{ Cat}^{\text{Span}_1(\mathcal{C})})$ by applying the enriched $\mathcal{V}$ -presheaf functor $\mathcal{P}_{\mathcal{V}}$.

- Since $\mathcal{V} \cong \mathcal{P}_{\mathcal{V}}(\text{B } 1_{\mathcal{V}})$ is $\mathcal{V}$ -atomically generated by the unit $1_{\mathcal{V}}$, and $1 : \mathcal{V} \rightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))$ sends the atomic generator $1_{\mathcal{V}}$ to the atomic object $* \in \text{Span}_n^{\text{enr}}(\mathcal{C}) \subset \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))$, thus is an internal left adjoint by [32, proposition 3.32].

□

Notation 3.1.4. Let $\Gamma : \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C})) \rightarrow \mathcal{V}$ be the right adjoint of $1 : \mathcal{V} \rightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))$.

Remark 3.1.5. • By construction, $\Gamma : \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C})) \rightarrow \mathcal{V}$ is the enriched Hom functor $\underline{\text{Hom}}_{\mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))}(*, -)$ where $*$ is the unit of $\text{Span}_n^{\text{enr}}(\mathcal{C}) \in \text{CAlg}(\text{Cat}^{\mathcal{V}})$, at the level of underlying space of objects, it is the final object $* \in \mathcal{C}$.

- Moreover, since the enriched Yoneda embedding $\mathcal{Y}_{\mathcal{V}}$ is $\mathcal{V}$ -fully faithful, for $x \in \text{Span}_n^{\text{enr}}(\mathcal{C}) \hookrightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))$, $\Gamma(x) = \underline{\text{Hom}}_{\mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))}(*, x) \cong \underline{\text{Hom}}_{\text{Span}_n^{\text{enr}}(\mathcal{C})}(*, x)$. Recall that $\underline{\text{Hom}}_{\text{Span}_n^{\text{enr}}(\mathcal{C})}(*, x) \cong \text{Span}_{n-1}(\mathcal{C}_{/x})$, and is enhanced into an object in $(n-1) \text{ Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})} \rightarrow (n-1) \text{ Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))} = \mathcal{V}$ by the $\text{B}^{n-2} \text{Span}_1(\mathcal{C})$ -action on $\text{Span}_{n-1}(\mathcal{C}_{/x})$, through the symmetric monoidal $(\infty, n-1)$ functor $\text{B}^{n-2} \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_{n-1}(\mathcal{C}) \xrightarrow{(-) \times x} \text{Span}_{n-1}(\mathcal{C}_{/x})$.

In conclusion:

Corollary 3.1.6. *There exists a chain of morphisms in $\text{CAlg}(\widehat{\text{Cat}^{\mathcal{V}}})$:*

$$\text{Span}_n^{\text{enr}}(\mathcal{C}) \xrightarrow{\mathcal{Y}_{\mathcal{V}}} \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C})) \xleftarrow{1} \mathcal{V}$$

, where $1 : \mathcal{V} \rightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C}))$ is equipped with a right adjoint Γ inside the non-full subcategory $\text{LMod}_{\mathcal{V}}(\text{Pr}^L) \rightarrow \widehat{\text{Cat}^{\mathcal{V}}}$, and $\text{Span}_n^{\text{enr}}(\mathcal{C})$ lands in the full subcategory $n \text{ Cat}^{\text{Span}_n(\mathcal{C})} \hookrightarrow \text{Cat}^{\mathcal{V}}$.

3.2 A lax symmetric monoidal (∞, n) functor

In this subsection, we again denote the presentably symmetric monoidal category $(n-1) \text{ Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$ as $\mathcal{V}$.

Construction 3.2.1. Consider the canonical symmetric adjunction

$$F : \mathbf{Spc} \rightleftarrows \mathcal{P}(\mathbf{Span}_1(\mathcal{C})) : G$$

where F is the unique symmetric monoidal colimit-preserving functor from $\mathbf{Spc}$ to $\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))$. Then there exists a symmetric monoidal adjunction

$$F_{n-1,*} : \mathbf{Cat}_{n-1} \rightleftarrows (n-1) \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))} : G_{n-1,*}$$

especially, $G_{n-1,*}$ can be enhanced into a lax symmetric monoidal functor.

Moreover, one can show that the above adjunction can be enhanced into an adjunction in $\widehat{\mathbf{Cat}}_n$: Since $F_* : \mathbf{Cat}_{n-1} \rightarrow (n-1) \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))}$ is a symmetric monoidal colimit-preserving functor, it gives rise to a morphism $[1] \rightarrow \mathbf{CAlg}(\mathbf{LMod}_{\mathbf{Cat}_{n-1}}(\mathbf{Pr}^L))$. Consider the lax symmetric monoidal inclusion $\mathbf{LMod}_{\mathbf{Cat}_{n-1}}(\mathbf{Pr}^L) \rightarrow \widehat{\mathbf{Cat}}_n$, which preserves adjunction by [6, corollary 4.19], thus we get a symmetric monoidal (∞, n) functor:

$$F_* : \mathbf{Cat}_{n-1} \rightarrow (n-1) \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))}$$

in $\widehat{\mathbf{Cat}}_n$ and the $(\infty, 1)$ right adjoint $G_* : (n-1) \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))} \rightarrow \mathbf{Cat}_{n-1}$ can be enhanced into a lax symmetric monoidal (∞, n) functor.

Here are some remarks one can make on the above construction:

Remark 3.2.2. • By the above construction, we enhance the presentably symmetric monoidal category $(n-1) \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))}$ into a symmetric monoidal (∞, n) category through the lax symmetric monoidal inclusion $\mathbf{LMod}_{\mathbf{Cat}_{n-1}}(\mathbf{Pr}^L) \rightarrow \widehat{\mathbf{Cat}}_n$, especially, by forgetting the $\mathbf{Cat}_{n-1}$ -action, one recover the presentably symmetric monoidal $(\infty, 1)$ category structure on $(n-1) \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))}$

$$\begin{array}{ccc} \mathbf{LMod}_{\mathbf{Cat}_{n-1}}(\mathbf{Pr}^L) & \longrightarrow & \widehat{\mathbf{Cat}}_n \\ \downarrow & & \downarrow \\ \mathbf{Pr}^L & \longrightarrow & \widehat{\mathbf{Cat}} \end{array}$$

- By replacing n with $n+1$, the forgetful functor

$$G_{n,*} : n \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))} \rightarrow \mathbf{Cat}_n$$

can be enhanced into a lax symmetric monoidal $\mathbf{Cat}_{n-1}$ -enriched functor (that is, a lax symmetric monoidal (∞, n) functor), thus preserving adjunction.

- By the corollary 5.3.6, there is a commutative diagram in $\mathbf{Op}$:

$$\begin{array}{ccc} \mathbf{LMod}_*(\mathbf{Cat}) & \xrightarrow{\cong} & \mathbf{Cat}^{\mathcal{P}(\ast)} \cong \mathbf{Cat} \\ \text{forget} \uparrow & & \uparrow G_* \\ \mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat}) & \longrightarrow & \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))} \end{array}$$

, thus for $\text{Span}_n^{\text{enr}}(\mathcal{C}) \in \text{CAlg}(\text{Cat}^{(n-1)\text{Cat}^\mathcal{V}})$, it recovers the symmetric monoidal (∞, n) -category $\text{Span}_n(\mathcal{C}) \in \text{CAlg}(\text{Cat}_n)$ under the transfer of enrichment along $G_{n-1,*} : (n-1)\text{Cat}^\mathcal{V} \rightarrow \text{Cat}_{n-1}$.

Now consider the following morphisms [corollary 3.1.6](#) in $\text{CAlg}(\text{Cat}^\mathcal{V})$, where $\mathcal{V} = (n-1)\text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$

$$\text{Span}_n^{\text{enr}}(\mathcal{C}) \xrightarrow{\mathfrak{L}_\mathcal{V}} \mathcal{P}_\mathcal{V}(\text{Span}_n^{\text{enr}}(\mathcal{C})) \xleftarrow{1} \mathcal{V}$$

Under the lax symmetric monoidal (∞, n) functor $G_{n,*} : \text{CAlg}(n\text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}) = \text{CAlg}(\text{Cat}^\mathcal{V}) \rightarrow \text{CAlg}(\text{Cat}_n)$, there are morphisms:

$$\text{Span}_n(\mathcal{C}) \xrightarrow{\mathfrak{L}_\mathcal{V}} \mathcal{P}_\mathcal{V}(\text{Span}_n^{\text{enr}}(\mathcal{C})) \xleftarrow{1} \mathcal{V}$$

where $\mathfrak{L}_\mathcal{V}$ is a symmetric monoidal (∞, n) functor and $1 : \mathcal{V} \rightarrow \mathcal{P}_\mathcal{V}(\text{Span}_n^{\text{enr}}(\mathcal{C}))$ admits the right adjoint Γ in Cat_n since $G_{n,*}$ preserves adjunction.

Remark 3.2.3. A priori, $\Gamma : \mathcal{P}_\mathcal{V}(\text{Span}_n^{\text{enr}}(\mathcal{C})) \rightarrow \mathcal{V}$ is a left $\mathcal{V}$ -module functor in $\text{LMod}_\mathcal{V}(\text{Pr}^L)$, thus is automatically a $\mathcal{V}$ -enriched functor. On the other hand, by the forgetful functor $G_{n-1,*} : \mathcal{V} \rightarrow \text{Cat}_{n-1}$, Γ can be regarded as an (∞, n) functor.

Thus Γ can be enhanced into a lax symmetric monoidal (∞, n) functor since 1 is a symmetric monoidal (∞, n) functor. And there is a chain of lax symmetric monoidal (∞, n) functors:

$$\text{Span}_n(\mathcal{C}) \rightarrow \mathcal{P}_\mathcal{V}(\text{Span}_n^{\text{enr}}(\mathcal{C})) \xrightarrow{\Gamma} \mathcal{V} = (n-1)\text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$$

Remark 3.2.4. One can see that: for $\Gamma(x) = \underline{\text{Hom}}_{\text{Span}_n^{\text{enr}}(\mathcal{C})}(*, x) \in n\text{Cat}^{\text{Span}_1(\mathcal{C})} \cong (n-1)\text{Cat}^{\text{Cat}^{\text{Span}_1(\mathcal{C})}}$, it can be regarded as the image of $\text{Span}_{n-1}^{\text{enr}}(\mathcal{C}_{/x}) \in (n-1)\text{Cat}^{\text{Span}_1(\mathcal{C}_{/x})}$ (by [corollary 2.6.7](#)) under the transfer of enrichment along the forgetful functor of slice category $\text{Span}_1(\mathcal{C}_{/x}) \rightarrow \text{Span}_1(\mathcal{C})$:

- Recall the diagram $\square$ in the proof of [lemma 2.6.4](#). Combined with the commutative diagram in [corollary 5.3.6](#), we have:

$$\begin{array}{ccccc} \text{LMod}_{\text{Span}_1(\mathcal{C}_{/x})}(\text{Cat})_{\text{closed}} & \hookrightarrow & \text{LMod}_{\text{Span}_1(\mathcal{C}_{/x})}(\text{Cat}) & \longrightarrow & \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}_{/x}))} \\ \downarrow & & \downarrow & & \downarrow g_* \\ \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}} & \hookrightarrow & \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}) & \longrightarrow & \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))} \end{array}$$

, where the middle vertical map is the forgetful functor along $(-) \times x : \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_1(\mathcal{C}_{/x})$, and g_* is the right adjoint of the transfer of enrichment along $(-) \times t : \mathcal{P}(\text{Span}_1(\mathcal{C})) \rightarrow \mathcal{P}(\text{Span}_1(\mathcal{C}_{/x}))$. Noticing the biadjunction:

$$(-) \times t : \text{Span}_1(\mathcal{C}) \rightleftarrows \text{Span}_1(\mathcal{C}_{/x}) : \text{forget along } x \rightarrow *$$

, taking the presheaf category gives rise to the adjunction

$$(-) \times t : \mathcal{P}(\text{Span}_1(\mathcal{C})) \rightleftarrows \mathcal{P}(\text{Span}_1(\mathcal{C}_{/x})) : \text{forget along } x \rightarrow *$$

. Thus, g_* is the transfer of enrichment along the forgetful functor $\mathcal{P}(\text{Span}_1(\mathcal{C}_{/x})) \rightarrow \mathcal{P}(\text{Span}_1(\mathcal{C}))$, and it restricts to $\text{Cat}^{\text{Span}_1(\mathcal{C}_{/x})} \rightarrow \text{Cat}^{\text{Span}_1(\mathcal{C})}$.

Notation 3.2.5. By the above remark, without abuse of notation, we again denote the image of $\text{Span}_{n-1}^{\text{enr}}(\mathcal{C}_{/x}) \in (n-1) \text{Cat}^{\text{Span}_1(\mathcal{C}_{/x})}$ under the transfer of enrichment along forgetful functor $\text{Span}_1(\mathcal{C}_{/x}) \rightarrow \text{Span}_1(\mathcal{C})$ again as $\text{Span}_n^{\text{enr}}(\mathcal{C}_{/x})$

Corollary 3.2.6. *There exists a lax symmetric monoidal (∞, n) functor:*

$$\text{Span}_n(\mathcal{C}) \rightarrow (n-1) \text{Cat}^{\text{Span}_1(\mathcal{C})}$$

assigning $x \in \text{Span}_n(\mathcal{C})$ with $\text{Span}_{n-1}^{\text{enr}}(\mathcal{C}_{/x})$ in [notation 3.2.5](#).

4 2 Shv(BG)

In this section, we fix $\mathcal{C} = \text{Stk}$, which is the category of Artin stacks of finite type with affine stabilizer over $\overline{k}_q$ where $q = p^n$ for a prime number p , and G is a smooth algebraic group over $\overline{k}_q$.

4.1 2 Shv(-)

Given a sheaf theory, which is a lax symmetric monoidal functor $\text{Shv}(-) : \text{Span}_1(\text{Stk}) \rightarrow 1 \text{ Pr}$,

- There is an induced colimit-preserving lax symmetric monoidal functor

$$\mathcal{P}(\text{Span}_1(\mathcal{C})) \rightarrow 1 \text{ Pr}$$

- There is an induced symmetric monoidal functor

$$\text{Env}(\text{Span}_1(\mathcal{C})) \rightarrow 1 \text{ Pr}$$

- There exists a localization:

$$l : \text{Env}(\mathcal{C}) \rightleftarrows \mathcal{C} : i$$

, which induces a localization:

$$\mathcal{P}(l) : \mathcal{P}(\text{Env}(\mathcal{C})) \rightleftarrows \mathcal{P}(\mathcal{C}) : \mathcal{P}(i)$$

, which is a symmetric monoidal adjunction, thus there is an induced symmetric monoidal adjunction by transfer of enrichment:

$$\mathcal{P}(l)_* : \text{Cat}^{\mathcal{P}(\text{Env}(\mathcal{C}))} \rightleftarrows \text{Cat}^{\mathcal{P}(\mathcal{C})} : \mathcal{P}(i)_*$$

.

- Moreover, we can enhance the symmetric monoidal left adjoint $\mathcal{P}(l)_* : \text{Cat}^{\mathcal{P}(\text{Env}(\mathcal{C}))} \rightarrow \text{Cat}^{\mathcal{P}(\mathcal{C})}$ into a symmetric monoidal $(\infty, 2)$ functor: The composition of morphisms $* \rightarrow \text{Env}(\mathcal{C}) \rightarrow \mathcal{C}$ in $\text{CAlg}(\text{Cat})$ induces a composition of morphisms $\mathcal{P}(*) \cong \text{Spc} \rightarrow \mathcal{P}(\text{Env}(\mathcal{C})) \rightarrow \mathcal{P}(\mathcal{C})$ in $\text{CAlg}(\text{Pr}^L)$; thus, there is an induced composition of morphisms $\text{Cat}^{\text{Spc}} \cong \text{Cat} \rightarrow \text{Cat}^{\mathcal{P}(\text{Env}(\mathcal{C}))} \rightarrow \text{Cat}^{\mathcal{P}(\mathcal{C})}$ in $\text{CAlg}(\text{Pr}^L)$; thus, $\text{Cat}^{\mathcal{P}(\text{Env}(\mathcal{C}))} \rightarrow \text{Cat}^{\mathcal{P}(\mathcal{C})}$ is a morphism in $\text{CAlg}(\text{LMod}_{\text{Cat}}(\text{Pr}^L)) \rightarrow \text{CAlg}(\text{Cat}_2)$.

- Thus, the right adjoint $\mathcal{P}(i)_* : \text{Cat}^{\mathcal{P}(\mathcal{C})} \rightarrow \text{Cat}^{\mathcal{P}(\text{Env}(\mathcal{C}))}$ can be enhanced into a lax symmetric monoidal $(\infty, 2)$ functor.
- Likewise, the composition of morphisms $* \rightarrow \text{Env}(\mathcal{C}) \rightarrow 1 \text{ Pr}$ in $\text{CAlg}(\text{Cat})$ induces $\text{Spc} \rightarrow \mathcal{P}(\text{Env}(\mathcal{C})) \rightarrow 1 \text{ Pr}$ in $\text{CAlg}(\text{Pr}^L)$, and thus $\text{Cat}^{\mathcal{P}(\text{Env}(\mathcal{C}))} \rightarrow \text{Cat}(1 \text{ Pr})$ is a morphism in $\text{CAlg}(\text{Cat}_2)$.
- Thus there is a lax symmetric monoidal $(\infty, 2)$ functor $\text{Cat}^{\text{Span}_1(\mathcal{C})} \rightarrow \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))} \rightarrow \text{Cat}^{\mathcal{P}(\text{Env}(\mathcal{C}))} \rightarrow \text{Cat}^{1 \text{ Pr}}$.

And by composition with the lax symmetric monoidal (∞, n) functor constructed in [corollary 3.2.6](#):

$$\text{Span}_2(\mathcal{C}) \rightarrow \text{Cat}^{\text{Span}_1(\mathcal{C})}$$

, there is a lax symmetric monoidal $(\infty, 2)$ functor

$$\text{Span}_2(\mathcal{C}) \rightarrow \text{Cat}^{1 \text{ Pr}}$$

.

Definition 4.1.1. Let $2 \text{ Shv}(-)$ denote the lax symmetric monoidal $(\infty, 2)$ functor:

$$2 \text{ Shv}(-) : \text{Span}_2(\mathcal{C}) \rightarrow \text{Cat}^{1 \text{ Pr}} \xrightarrow{\mathcal{P}_{1 \text{ Pr}}(-)} \text{LMod}_{1 \text{ Pr}}(1 \text{ Pr}^L) \cong 2 \text{ Pr}$$

where $\mathcal{P}_{1 \text{ Pr}}(-)$ is the enriched presheaf functor.

Remark 4.1.2. Noticing the following commutative diagram:

$$\begin{array}{ccccc} \text{Cat}^{1 \text{ Pr}} & \xrightarrow{\mathcal{P}_{1 \text{ Pr}}(-)} & \text{LMod}_{1 \text{ Pr}}(1 \text{ Pr}) & \longrightarrow & \text{LMod}_{1 \text{ Pr}}(\text{Pr}^L) \\ (\text{Spc} \rightarrow 1 \text{ Pr})_* \uparrow & & \uparrow (-) \otimes 1 \text{ Pr} & & \uparrow (-) \otimes 1 \text{ Pr} \\ \text{Cat}^{\text{Spc}} \cong \text{Cat} & \xrightarrow{\mathcal{P}(-)} & 1 \text{ Pr} & \longrightarrow & \text{Pr}^L \end{array}$$

Especially, we can extract the following diagram in $\text{CAlg}(\text{Pr}^L)$:

$$\begin{array}{ccc} \text{Cat}^{1 \text{ Pr}} & \xrightarrow{\mathcal{P}_{1 \text{ Pr}}(-)} & \text{LMod}_{1 \text{ Pr}}(1 \text{ Pr}) \\ (\text{Spc} \rightarrow 1 \text{ Pr})_* \uparrow & & \uparrow (-) \otimes 1 \text{ Pr} \\ \text{Cat}^{\text{Spc}} \cong \text{Cat} & \xrightarrow{\mathcal{P}(-)} & 1 \text{ Pr} \end{array}$$

where $\mathcal{P}_{1 \text{ Pr}}$ preserves colimits by [32, proposition 6.40]. Thus $\mathcal{P}_{1 \text{ Pr}}(-) : \text{Cat}^{1 \text{ Pr}} \rightarrow \text{LMod}_{1 \text{ Pr}}(1 \text{ Pr})$ can be enhanced into a symmetric monoidal $(\infty, 2)$ functor.

Remark 4.1.3. As a supplementary remark, for $\mathcal{V} \in \text{Pr}^\kappa$, $\mathcal{P}_{\mathcal{V}}(-) : \text{Cat}^{\mathcal{V}} \rightarrow \text{LMod}_{\mathcal{V}}(\text{Pr}^L)$ factors through the non-full subcategory $\text{LMod}_{\mathcal{V}}(\text{Pr}_\kappa^L)$: by construction of the enriched presheaf category in [16], $\mathcal{P}_{\mathcal{V}}(\mathcal{M})$ is κ -compactly generated by $v \otimes m$ where $v \in \mathcal{V}^\kappa$ and $m \in \mathcal{M}$, since there is functorial equivalence

$$\text{Map}_{\mathcal{P}_{\mathcal{V}}(\mathcal{M})}(v \otimes m, -) \cong \text{Map}_{\mathcal{V}}(v, \underline{\text{Hom}}_{\mathcal{P}_{\mathcal{V}}(\mathcal{M})}(m, -))$$

and m is $\mathcal{V}$ -atomic in $\mathcal{P}_{\mathcal{V}}(\mathcal{M})$. Moreover, the action map $\mathcal{V} \otimes \mathcal{P}_{\mathcal{V}}(\mathcal{M})$ preserves κ -compact objects, thus the enriched presheaf category of $\mathcal{M} \in \text{Cat}^{\mathcal{V}}$ lands in $\text{LMod}_{\mathcal{V}}(\text{Pr}_{\kappa}^L)$. By the description of κ -compact generators in $\mathcal{P}_{\mathcal{V}}(\mathcal{M})$, for any $F : \mathcal{M} \rightarrow \mathcal{N}$ in $\text{Cat}^{\mathcal{V}}$, the induced morphism $\mathcal{P}_{\mathcal{V}}(\mathcal{M}) \rightarrow \mathcal{P}_{\mathcal{V}}(\mathcal{N})$ in $\text{LMod}_{\mathcal{V}}(\text{Pr}^L)$ lands in $\text{LMod}_{\mathcal{V}}(\text{Pr}_{\kappa}^L)$.

Remark 4.1.4. Moreover, the $(\infty, 2)$ structure on 2Pr can be described as follows: given $\mathcal{M}, \mathcal{N} \in \text{LMod}_{1\text{Pr}}(1\text{Pr})$, the Hom category $\underline{\text{Hom}}_{2\text{Pr}}(\mathcal{M}, \mathcal{N}) \cong \text{Fun}_{(1\text{Pr})^{\aleph_1}}^{\aleph_1}(\mathcal{M}^{\aleph_1}, \mathcal{N}^{\aleph_1})$, that is, the category of $\aleph_1$ -continuous $(1\text{Pr})^{\aleph_1}$ -linear functors between $\mathcal{M}^{\aleph_1}$ and $\mathcal{N}^{\aleph_1}$.

Remark 4.1.5. Noticing the following symmetric monoidal adjunction of (large version) adjoining the $\aleph_1$ -filtered colimits:

$$\widehat{\text{Cat}} \rightleftarrows \widehat{\text{Cat}}^{\aleph_1}$$

where $\widehat{\text{Cat}}^{\aleph_1}$ is the very large category of large categories with small $\aleph_1$ -filtered colimits, and functors preserving small $\aleph_1$ -filtered colimits. Especially, the left adjoint restricts to the symmetric monoidal functor $\text{Ind}_{\aleph_1}(-) : \text{Cat} \rightarrow \widehat{\text{Cat}}^{\aleph_1}$; then there exists an induced symmetric monoidal adjunction:

$$\widehat{\text{Cat}}^{\widehat{\text{Cat}}} \rightleftarrows \widehat{\text{Cat}}^{\widehat{\text{Cat}}^{\aleph_1}}$$

, and the unit map valued at $2\text{Pr} \in \widehat{\text{Cat}}_2$ gives a map

$$2\text{Pr} \rightarrow 2\text{Pr}^{\sharp}$$

in $\text{CAlg}(\widehat{\text{Cat}}^{\widehat{\text{Cat}}}) \cong \widehat{\text{Cat}}_2$, where it changes the Hom category $\text{Fun}_{(1\text{Pr})^{\aleph_1}}^{\aleph_1}(\mathcal{M}^{\aleph_1}, \mathcal{N}^{\aleph_1})$ between $\mathcal{M}, \mathcal{N}$ in 2Pr to $\text{Ind}_{\aleph_1}(\text{Fun}_{(1\text{Pr})^{\aleph_1}}^{\aleph_1}(\mathcal{M}^{\aleph_1}, \mathcal{N}^{\aleph_1}))$, which is the internal Hom of the Cartesian closed symmetric monoidal 2Pr .

Remark 4.1.6. 4.2 $2\text{Shv}(\text{BG})$

In this subsection, we fix our sheaf theory as the one developed in [22], which is a lax symmetric monoidal functor:

$$\text{Shv}(-) : \text{Span}_1(\mathcal{C}) \rightarrow \text{Pr}^L$$

assigning a stack $X \in \text{Stk}$ with the ∞ -category of l -adic sheaves $\text{Shv}(X)$ on X , and given a span

$$x \xleftarrow{f} z \xrightarrow{g} y \quad \text{it sends } f \text{ to } f^* : \text{Shv}(z) \rightarrow \text{Shv}(x), \text{ and } g \text{ to } g_! : \text{Shv}(z) \rightarrow \text{Shv}(y).$$

Remark 4.2.1. The l -adic sheaf theory we take $\text{Shv}(-) : \text{Span}(\text{Stk}) \rightarrow \text{Pr}^L$ lands inside the non-full subcategory 1Pr :

- For $\mathcal{X} \in \text{Stk}$ with atlas of a scheme $j : X \rightarrow \mathcal{X}$, $\text{Shv}(\mathcal{X})$ is compactly generated by objects in the form of $j_!(\mathcal{F})$ for compact object $\mathcal{F} \in \text{Shv}(X)^c$.
- For a morphism $f : \mathcal{X} \rightarrow \mathcal{Y}$ in Stk , $f_!$ preserves compactness since $f^!$ is continuous.
- However, f^* doesn't preserve compactness in general (for example, $\frac{1}{\text{BG}_n}$ is not compact). Instead, f^* preserves constructibility (see its definition in [20, definition 2.1.9]), and a constructible sheaf on any $\mathcal{X} \in \mathcal{C}$ is $\aleph_1$ -compact. Thus, for an atlas $h : Y \rightarrow \mathcal{Y}$ and

compact $\mathcal{F} \in \mathrm{Shv}(Y)^c$, we take the $\aleph_1$ -compact generator $h_! \mathcal{F}$ of $\mathcal{Y}$, and by base change, we have $f^* h_! (\mathcal{F}) \cong h'_! (f')^* (\mathcal{F})$

$$\begin{array}{ccc} Y \times_{\mathcal{Y}} \mathcal{X} & \xrightarrow{f'} & Y \\ h' \downarrow & & \downarrow h \\ \mathcal{X} & \xrightarrow{f} & \mathcal{Y} \end{array}$$

. Moreover, $(f')^* (\mathcal{F})$ is constructible, thus $\aleph_1$ -compact, and so is $h'_! (f')^* (\mathcal{F})$.

Definition 4.2.2. Given a sheaf theory $\mathrm{Shv}(-)$ and a morphism $f : x \rightarrow y$ in $\mathcal{C}$, with Čech nerve $x_{\bullet} \rightarrow y$ we say that f satisfies Shv-descent if $\mathrm{Shv}(y)$ is the colimit of $\mathrm{Shv}(x_{\bullet})$ under the lower $!$ transition map. If the pullback of f satisfies Shv-descent, we say f satisfies universal Shv-descent.

Example 4.2.3. Consider the étale sheaf theory $\mathrm{Shv}(-)$ developed in [22] and a morphism $f : * \rightarrow \mathrm{BG}$, then we have $\mathrm{Shv}(\mathrm{BG}) \cong \mathrm{Tot}(\mathrm{Shv}(*) \rightarrow \mathrm{Shv}(G) \cdots)$ can be computed as either upper $*$ or upper $!$, which differs by a shift. And the limit is equivalent to the colimit taken in 1 Pr by passing the right adjoint upper $!$ to left adjoint lower $!$. Thus $f : * \rightarrow \mathrm{BG}$ satisfies Shv-descent. Likewise, the above argument works for any pullback of $* \rightarrow \mathrm{BG}$, thus $f : * \rightarrow \mathrm{BG}$ satisfies universal Shv-descent.

Proposition 4.2.4. Given a morphism $f : x \rightarrow y \in \mathcal{C}$, if f satisfies universal Shv-descent, then $2 \mathrm{Shv}(y) \cong \mathrm{Tot}(2 \mathrm{Shv}(x_{\bullet}))$.

Proof. This is [37, proposition 4.2]. □

Thus we have the following diagram in $\mathrm{LMod}_{1 \mathrm{Pr}}(\mathrm{Pr}^L)$.

$$2 \mathrm{Shv}(\mathrm{BG}) \xrightarrow{\cong} \mathrm{Tot}(2 \mathrm{Shv}(*) \longrightarrow 2 \mathrm{Shv}(G) \rightrightarrows 2 \mathrm{Shv}(G^2) \rightrightarrows \cdots)$$

Recall the following proposition on descent and monadicity:

Proposition 4.2.5. Given a diagram $F : J \rightarrow \mathrm{Cat}_1$, assume that there exists an object $0 \in J$ such that for any object $x \in J$, there exists a morphism $0 \rightarrow x \in J$, and the structure morphism $s : \lim F \rightarrow F(0)$ admits a left adjoint L , then the adjunction

$$L : F(0) \rightleftarrows \lim F : s$$

is monadic.

Proof. This is [25, proposition 4.5.7] □

Corollary 4.2.6. The adjunction induced by the atlas map $f : * \rightarrow \mathrm{BG}$:

$$f^* : 2 \mathrm{Shv}(\mathrm{BG}) \rightleftarrows 2 \mathrm{Shv}(*) : f_*$$

is monadic, and the monad is the group object $G \in \mathcal{C}$. Thus there is an equivalence $2 \mathrm{Shv}(\mathrm{BG}) \cong \mathrm{LMod}_G(2 \mathrm{Shv}(*))$

Proof. • By construction, $2 \operatorname{Shv}(-) : \operatorname{Span}_2(\mathcal{C}) \rightarrow \operatorname{LMod}_{1 \operatorname{Pr}}(\operatorname{Pr}^L)$ gives us a biadjunction

$$f^* : 2 \operatorname{Shv}(\operatorname{BG}) \rightleftarrows 2 \operatorname{Shv}(*) : f_*$$

where f^* sends an object $x \rightarrow \operatorname{BG} \in 2 \operatorname{Shv}(\operatorname{BG})$ to $x \times_{\operatorname{BG}} *$ and f_* sends an object $y \rightarrow * \in 2 \operatorname{Shv}(*)$ to $y \rightarrow * \rightarrow \operatorname{BG} \in 2 \operatorname{Shv}(\operatorname{BG})$.

- By the above proposition, our adjunction is monadic.
- Moreover, consider the commutative diagram $* \rightarrow \operatorname{BG} \rightarrow *$, which induces an equivalence:

$$\begin{array}{ccc} \operatorname{BG} & \xleftarrow{f} & * \\ & \searrow s & \swarrow \\ & * & \end{array}$$

$$\begin{array}{ccc} 2 \operatorname{Shv}(\operatorname{BG}) & \xrightarrow{f^*} & 2 \operatorname{Shv}(*) \\ & \searrow s^* & \swarrow \\ & 2 \operatorname{Shv}(*) & \end{array}$$

where $s : \operatorname{BG} \rightarrow *$ is the structure map.

- The above commutative diagram makes $2 \operatorname{Shv}(\operatorname{BG})$ as a $2 \operatorname{Shv}(*)$ -left module category, and the functor f^* is thus enhanced into a $2 \operatorname{Shv}(*)$ -left module functor. Thus the right adjoint f_* (noticing that there we have a biadjunction) is naturally enhanced into a lax $2 \operatorname{Shv}(*)$ -left module functor, which is in fact strict: since both f_* and f^* are colimit-preserving, thus it suffices to check the generators $y, x \rightarrow *$, which is obviously true.
- Thus, the monad is represented by an object $T \in \operatorname{Alg}(2 \operatorname{Shv}(*)$, the underlying object of T is just $f^* f_*(*) \cong G \in 2 \operatorname{Shv}(*)$. And the algebra object structure on T is given by $T \times T \cong f^*(f_* f^*) f_*(*) \rightarrow f^* \operatorname{id} f_*(*) \cong T$, which is the multiplication map for the group scheme G .

□

5 Appendix on enriched category theory

5.1 Enriched category theory

In this subsection, we review two different models of enriched category theory; one is categorical algebra by [10], and the other is a weakly left tensored category by [15]. These two models are equivalent, as indicated by [15].

Notation 5.1.1. Let $\operatorname{Ass} := \Delta^{op}$ be the category of finite non-empty totally ordered sets and order preserving maps. We call a map $[n] \rightarrow [m] \in \operatorname{Ass}$

- inert if it corresponds to a map of Δ in the form $[m] \cong \{i, i+1, \dots, i+m\} \subset [n]$ for some $i \geq 0$.
- standard inert if it is inert and $m = 1$.

- active if it corresponds to a map of Δ that preserves the minimum and maximum.

The following definitions follow [10, subsection 4] and [15, subsection 4.1].

Definition 5.1.2. Given a space $X \in \mathbf{Spc}$, let $\text{Ass}_X \rightarrow \text{Ass}$ be the coCartesian fibration classifying the functor $i_* X \in \text{Fun}(\text{Ass}, \text{Cat})$ where $i_* : \text{Cat} \rightarrow \text{Fun}(\text{Ass}, \text{Cat})$ is the right Kan extension along $[0] : * \rightarrow \text{Ass}$.

Definition 5.1.3. A categorical algebra in a non-symmetric operad $\mathcal{V}^\otimes \rightarrow \text{Ass}$ with underlying space of objects X is a morphism $\text{Ass}_X^{\text{op}} \rightarrow \mathcal{V}^\otimes$ over Ass preserving the coCartesian liftings over the inert maps in Ass .

One can show that $\text{Ass}_X \rightarrow \text{Ass}$ is a generalized non-symmetric operad defined in [10, 2.4]. Moreover, the underlying category of Ass_X is $(\text{Ass}_X)_{[1]} \cong X \times X$, and given a categorical algebra $\mathcal{C} : \text{Ass}_X \rightarrow \mathcal{V}^\otimes$, the induced morphism over $[1] \in \text{Ass}$ is a functor $\mathcal{C}(-, -) : X \times X \rightarrow \mathcal{V}$, which can be regarded as the enriched Hom of $\mathcal{C}$, assigning for any objects $x, y \in X$ an object $\mathcal{C}(x, y) \in \mathcal{V}$.

Thus, categorical algebra can be regarded as a prototype of an enriched category, and given a categorical algebra $\mathcal{C} : \text{Ass}_X \rightarrow \mathcal{V}^\otimes$, we call an object x in the space X an object in $\mathcal{C}$, also denoted as $x \in \mathcal{C}$.

And one can assemble $\text{Alg}_{\text{Ass}_X}(\mathcal{V})$ for all $X \in \mathbf{Spc}$ and all $\mathcal{V} \in$ together as a fibration over $\mathbf{Spc} \times \text{Op}^{ns}$:

Theorem 5.1.4. *There exists a map $\text{PreCat} \rightarrow \text{Op}^{ns} \times \mathbf{Spc}$, which is coCartesian fibration over Op^{ns} and Cartesian fibration over $\mathbf{Spc}$, and the fiber over $(\mathcal{V}^\otimes, X) \in \text{Op}^{ns} \times \mathbf{Spc}$ is the category $\text{Alg}_{\text{Ass}_X}(\mathcal{V})$ of categorical algebras in $\mathcal{V}$ with underlying space of objects X .*

Proof. This is [15, notation 4.40 and remark 4.41]. □

Notation 5.1.5. Given a morphism between ono-symmetric operad $f : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$, the coCartesian lifting of f in $\text{PreCat} \rightarrow \text{Op}^{ns}$ is called transfer of enrichment along f , and we denote it as:

$$f_*^{\text{cat}} : \text{Alg}_{\text{cat}}(\mathcal{V}) \rightarrow \text{Alg}_{\text{cat}}(\mathcal{W})$$

.

Notation 5.1.6. Given a non-symmetric operad $\mathcal{V}^\otimes \in \text{Op}^{ns}$, let $\text{Alg}_{\text{cat}}(\mathcal{V})$ denote the fiber product $\text{PreCat} \times_{\text{Op}^{ns}} \{\mathcal{V}^\otimes\}$, which is a Cartesian fibration over $\mathbf{Spc}$.

Definition 5.1.7. Given a monoidal category $\mathcal{V}^\otimes \in \text{Mon} \rightarrow \text{Op}^{ns}$, a morphism $\mathcal{C} \rightarrow \mathcal{D} \in \text{Alg}_{\text{cat}}(\mathcal{V})$ is $(\mathcal{V})$ -fully faithful if it's the Cartesian lifting of the Cartesian fibration $\text{Alg}_{\text{cat}}(\mathcal{V}) \rightarrow \mathbf{Spc}$.

Now we introduce another model of enriched category by weakly left tensored categories:

Definition 5.1.8. Let $\mathcal{V}^\otimes \rightarrow \text{Ass}$ be an ∞ -operad and $\phi : \mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ a map of coCartesian fibrations relative to the inert maps of Ass that preserves the maximum.

We call $\phi : \mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ an ∞ -category weakly left tensored over $\mathcal{V}$ (or an ∞ -category with weak left $\mathcal{V}$ -action) if the following conditions hold:

- (1) for every $n \geq 0$ the map $[0] \cong \{n\} \subset [n]$ in Δ induces an equivalence

$$\theta : \mathcal{M}_{[n]}^\otimes \rightarrow \mathcal{V}_{[n]}^\otimes \times \mathcal{M}_{[0]}^\otimes$$

- (2) for every $X, Y \in \mathcal{M}^\otimes$ lying over $[m], [n] \in \text{Ass}$ the coCartesian lift $Y \rightarrow Y'$ of the map $[0] \cong \{n\} \subset [n]$ in Δ induces a pullback square:

$$\begin{array}{ccc} \mathcal{M}^\otimes(X, Y) & \longrightarrow & \mathcal{M}^\otimes(X, Y') \\ \downarrow & & \downarrow \\ \mathcal{V}^\otimes(\phi(X), \phi(Y)) & \longrightarrow & \mathcal{V}^\otimes(\phi(X), \phi(Y')) \end{array}$$

Definition 5.1.9. Let $\phi : \mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes, \phi' : \mathcal{M}'^\otimes \rightarrow \mathcal{V}'^\otimes$ be ∞ -categories with weak left action. We call a commutative diagram square of ∞ -categories over Ass

$$\begin{array}{ccc} \mathcal{M}^\otimes & \xrightarrow{\gamma} & \mathcal{M}'^\otimes \\ \phi \downarrow & & \downarrow \phi' \\ \mathcal{V}^\otimes & \xrightarrow{\rho} & \mathcal{V}'^\otimes \end{array}$$

a lax linear functor between weakly left tensored ∞ -categories if γ preserves coCartesian lifts of inert morphisms of Ass that preserves the maximum.

Notation 5.1.10. Let $\text{Cat}_{\infty/\text{Ass}}^{\max} \subset \text{Cat}_{\infty/\text{Ass}}$ be the subcategory of coCartesian fibrations relative to the collection of inert morphisms which preserves the maximum. Let

$$\omega\text{LMod} \subset \text{Op}^{ns} \times_{\text{Fun}(\{1\}, \text{Cat}_{\infty/\text{Ass}}^{\max})} \text{Fun}([1], \text{Cat}_{\infty/\text{Ass}}^{\max})$$

be the full subcategories of ∞ -categories with a weak left action over some ∞ -operad.

Remark 5.1.11. By [15, proposition 3.33] and [19, proposition 2.72], there is an equivalence $_{/\text{LM}^\otimes} \cong \omega\text{LMod}$. This implies that the definition of weakly left tensored categories agrees with the one [25, definition 4.2.1.12]. Moreover, under the identification of ωLMod with $\text{Op}_{/\text{LM}^\otimes}$, there is a non-full subcategory inclusion $\text{LMod} \rightarrow \omega\text{LMod}$, which corresponds to weakly left tensored categories $\mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ which is a morphism between coCartesian fibrations over Ass , and lax linear functors (γ, ρ) which preserve all coCartesian liftings.

Theorem 5.1.12. [15, 3.32] *The canonical functor $\omega\text{LMod} \rightarrow \text{Op}^{ns}$ is a biCartesian fibration.*

Definition 5.1.13. Let $\phi : \mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ be a weakly left tensored ∞ -category.

- (1) We say that ϕ exhibits $\mathcal{M}$ as (left) pseudo-enriched in $\mathcal{V}$ if $\mathcal{V}^\otimes \rightarrow \text{Ass}$ is a monoidal ∞ -category and for every $X, Y \in \mathcal{M}$ and $V_1, \dots, V_n \in \mathcal{V}$, the map

$$\text{Mul}_{\mathcal{M}}(V_1 \otimes \dots \otimes V_n, X; Y) \rightarrow \text{Mul}_{\mathcal{M}}(V_1, \dots, V_n, X; Y)$$

induced by the active morphism $(V_1, \dots, V_n) \rightarrow V_1 \otimes \dots \otimes V_n$ is an equivalence.

- (2) a morphism object of $X, Y \in \mathcal{M}$ is an object $\text{Mor}_{\mathcal{M}}(X, Y) \in \mathcal{V}$ together with a multimorphism $\alpha \in \text{Mul}_{\mathcal{M}}(\text{Mor}_{\mathcal{M}}(X, Y), X; Y)$ that induces for all objects $Z_1, \dots, Z_n \in \mathcal{V}$ an equivalence of spaces:

$$\text{Mul}_{\mathcal{V}}(Z_1, \dots, Z_n; \text{Mor}_{\mathcal{M}}(X, Y)) \cong \text{Mul}_{\mathcal{M}}(Z_1, \dots, Z_n, X; Y)$$

is an equivalence.

- (3) We say that $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ exhibits $\mathcal{M}$ enriched in $\mathcal{V}$ if for every $X, Y \in \mathcal{M}$, there exists a morphism object $\text{Mor}_{\mathcal{M}}(X, Y)$.

Since we are mainly interested in the category enriched in a monoidal category, we restrict ourselves to the following notations:

Notation 5.1.14. (1) Let PLMod denote the non-full subcategory of ωLMod consisting of pseudo-

enriched categories and lax linear functors

$$\begin{array}{ccc} \mathcal{M}^{\otimes} & \xrightarrow{f} & \mathcal{N}^{\otimes} \\ \downarrow & & \downarrow \\ \mathcal{V}^{\otimes} & \xrightarrow{\alpha} & \mathcal{W}^{\otimes} \end{array} \quad \text{where } \alpha \text{ is a monoidal functor.}$$

- (2) Let ωPLMod be the full subcategory of ωLMod consisting of pseudo-enriched categories. Noticing that $\text{PLMod} \rightarrow \text{Mon}$ is the pullback of ωPLMod along the inclusion $\text{Mon} \rightarrow \text{Mon}^{\text{lax}}$ (where Mon^{lax} is the full subcategory of Op^{ns} consisting of monoidal categories).
- (3) Let Enr denote the full subcategory of ωPLMod consisting of enriched categories.
- (4) Fix $\mathcal{V} \in \text{Op}^{\text{ns}}$, let $\text{Cat}^{\mathcal{V}}$ denote the fiber of $\text{Enr} \rightarrow \text{Op}^{\text{ns}}$ over $\mathcal{V}$.
- (5) Fix $\mathcal{V} \in \text{Mon}$, let $\text{PLMod}_{\mathcal{V}}$ denote the fiber of $\text{PLMod} \rightarrow \text{Mon}$ over $\mathcal{V}$.

And there is a comparison functor by [15, theorem 6.7, proposition 6.10]:

Theorem 5.1.15. *There exists a fully faithful embedding $i : \text{Enr} \hookrightarrow \text{PreCat}$ over Op^{ns} that admits a left adjoint loc .*

Moreover, one can describe the essential image of i by the ‘complete categorical algebra’:

Definition 5.1.16. [10, proposition 5.42] Given a monoidal category $\mathcal{V}^{\otimes}$, a categorical algebra in $\mathcal{V}$ is called complete if it’s local with respect to the morphism $E^1 \rightarrow E^0$ in $\text{Alg}_{\text{cat}}(\mathcal{V})$, where E^1 is $\text{Ass}_{\{0,1\}} \rightarrow \text{Ass} \xrightarrow{1_{\mathcal{V}}} \mathcal{V}^{\otimes}$, and E^0 is $\text{Ass}_{\{0\}} \rightarrow \text{Ass} \xrightarrow{1_{\mathcal{V}}} \mathcal{V}^{\otimes}$, and the map $E^1 \rightarrow E^0$ is induced by $\{0, 1\} \rightarrow \{0\}$.

Theorem 5.1.17. *The essential image of $i : \text{Enr} \hookrightarrow \text{PreCat}$ over a monoidal category $\mathcal{V} \in \text{Mon} \rightarrow \text{Op}^{\text{ns}}$ is precisely the complete categorical algebras in $\mathcal{V}$.*

Definition 5.1.18. Given a monoidal category $\mathcal{V}$, and a morphism $f : \mathcal{C} \rightarrow \mathcal{D} \in \text{Alg}_{\text{cat}}(\mathcal{V})$, f is essential surjective if for every object $y \in \mathcal{D}$, there exists $x \in \mathcal{C}$ and a map $E^1 \rightarrow \mathcal{D}$ connecting the $f(x)$ and y in $\mathcal{D}$.

Remark 5.1.19. By [15, corollary 6.13] and [10, theorem 5.6.6], the fully faithful embedding $i_{\mathcal{V}} : \text{Cat}^{\mathcal{V}} \hookrightarrow \text{Alg}_{\text{cat}}(\mathcal{V})$ for a monoidal category $\mathcal{V}$ is a localization with respect to $\mathcal{V}$ -fully faithful and essential surjective morphisms.

5.2 Fibrations

For practical usage, we mainly consider the enriched category theory over $\text{Mon}^{lax} \hookrightarrow \text{Op}^{ns}$

Theorem 5.2.1. $\text{Enr} \times_{\text{Op}^{ns}} \text{Mon}^{lax} \rightarrow \text{Mon}^{lax}$ is a coCartesian fibration, and the left adjoint $\text{loc} : \text{PreCat} \times_{\text{Op}^{ns}} \text{Mon}^{lax} \rightarrow \text{Enr} \times_{\text{Op}^{ns}} \text{Mon}^{lax}$ preserves coCartesian edges.

Proof. By [10, lemma 5.7.5], the coCartesian lifting over a lax monoidal functor $f : \mathcal{V} \rightarrow \mathcal{W}$ in Mon^{lax} , which is the transfer of enrichment along f , preserves enriched fully faithful and essential surjective morphisms. Thus, by [25, lemma 2.2.4.11], $\text{Enr} \times_{\text{Op}^{ns}} \text{Mon}^{lax} \rightarrow \text{Mon}^{lax}$ is a coCartesian fibration, and L preserves the coCartesian edges. $\square$

Notation 5.2.2. Given a morphism $f : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$, we again call the coCartesian lifting over f the transfer of enrichment along f , denoted as

$$f_* : \text{Enr}_{\mathcal{V}} \rightarrow \text{Enr}_{\mathcal{W}}$$

.By construction, this is the composition of $\text{Cat}^{\mathcal{V}} \hookrightarrow \text{Alg}_{\text{cat}}(\mathcal{V}) \xrightarrow{f_*^{\text{cat}}} \text{Alg}_{\text{cat}}(\mathcal{W}) \xrightarrow{\text{loc}_{\mathcal{W}}} \text{Cat}^{\mathcal{W}}$ where the arrow in the middle is the transfer of enrichment along f for the categorical algebras.

Proposition 5.2.3. Given a monoidal functor $f : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$ in Mon , if it admits a right adjoint $g : \mathcal{W}^\otimes \rightarrow \mathcal{V}^\otimes$ in Mon^{lax} , then there exists an adjunction

$$f_* : \text{Cat}^{\mathcal{V}} \rightleftarrows \text{Cat}^{\mathcal{W}} : g_*$$

Proof. For the model of weakly left tensored categories, this is by [16, 4.70]. For the model of categorical algebras, this is by [10, 5.7.17]. $\square$

Proposition 5.2.4. Consider the Cartesian fibration ωLMod over Op , it restricts to a Cartesian fibration PLMod over Mon , and the inclusion $\text{PLMod} \rightarrow \omega\text{LMod}$ preserves the Cartesian edges.

Proof. It suffices to show the Cartesian edge in ωLMod at a point $N^\otimes \rightarrow B^\otimes \in \text{PLMod}$ remains inside PLMod .

By [16, 2.62], the Cartesian edge in ωLMod over a map of operads α is just the pullback of operads: Given two weakly left tensored categories $M^\otimes \rightarrow A^\otimes$ and $N^\otimes \rightarrow B^\otimes$, an morphism of weakly left tensored modules $\alpha^* : M^\otimes \rightarrow N^\otimes$ is Cartesian iff the following diagram is a pullback diagram:

$$\begin{array}{ccc} M^\otimes & \xrightarrow{\alpha^*} & N^\otimes \\ \downarrow & \lrcorner & \downarrow \\ A^\otimes & \xrightarrow{\alpha} & B^\otimes \end{array}$$

, that is, for any $a_1, a_2, \dots, a_n \in A$ and $m, n \in M$, the following map induced is an equivalence:

$$\text{Mul}_M(a_1, \dots, a_n, m; n) \rightarrow \text{Mul}_N(\alpha(a_1), \dots, \alpha(a_n), \alpha^*(m); \alpha^*(n))$$

Thus if $\alpha : A \rightarrow B$ is a monoidal functor, then if N is pseudo-enriched in A , and for any $a_1, a_2, \dots, a_n \in A$ and $m, n \in M$, we have the following commutative diagram:

$$\begin{array}{ccc}
\text{Mul}_M(a_1 \otimes \cdots \otimes a_n, m; n) & \xrightarrow{\cong} & \text{Mul}_N(\alpha(a_1 \otimes \cdots \otimes a_n), \alpha^*(m); \alpha^*(n)) \\
\downarrow & & \downarrow \\
\text{Mul}_M(a_1, \dots, a_n, m; n) & \xrightarrow{\cong} & \text{Mul}_N(\alpha(a_1), \dots, \alpha(a_n), \alpha^*(m); \alpha^*(n))
\end{array}$$

where the vertical arrows are induced by the active map corresponding to the tensor product. Thus the left vertical map is equivalence, and this implies M is pseudo-enriched in A . $\square$

Remark 5.2.5. The above statement is not true if the morphism $\alpha : A \rightarrow B$ is in Op instead of Mon , for example, by [16, corollary 4.22], the pullback along the lax monoidal functor $\mathcal{V} \rightarrow \text{Env}(\mathcal{V})$ takes the categories pseudo-enriched in $\text{Env}(\mathcal{V})$ to weakly left tensored categories over $\mathcal{V}$.

Proposition 5.2.6. *The (non-full) subcategory inclusion $\text{LMod} \rightarrow \text{PLMod}$ preserves Cartesian edges over Mon .*

Proof. We have already known $\text{LMod} \rightarrow \text{Mon}$ is a Cartesian fibration by [25, corollary 4.2.3.2], thus it suffices to show:

- Given a left tensored ∞ -category $\mathcal{N}^\otimes \rightarrow \mathcal{W}^\otimes$, and a monoidal functor $f : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$, the pullback in ωLMod :

$$\begin{array}{ccc}
\mathcal{M}^\otimes & \xrightarrow{f!} & \mathcal{N}^\otimes \\
\downarrow \lrcorner & & \downarrow \\
\mathcal{V}^\otimes & \xrightarrow{f} & \mathcal{W}^\otimes
\end{array}$$

lies in $\text{LMod} \subset \text{PLMod}$, which is true since pullback of coCartesian fibration is again coCartesian fibration.

- For any $\mathcal{N}' \in \text{LMod}_{\mathcal{V}}(\text{Cat})$, and a linear functor $\begin{array}{ccc} \mathcal{N}'^\otimes & \xrightarrow{F} & \mathcal{N}^\otimes \\ \downarrow & & \downarrow \\ \mathcal{V}^\otimes & \xrightarrow{f} & \mathcal{W}^\otimes \end{array}$, there is a unique

$\mathcal{V}$ -linear functor $G : \mathcal{N}' \rightarrow \mathcal{M}$ making the following diagram

$$\begin{array}{ccccc}
& & \mathcal{N}'^\otimes & & \\
& \searrow & \downarrow & \searrow F & \\
& & \mathcal{M}^\otimes & \xrightarrow{f!} & \mathcal{N}^\otimes \\
& \searrow G & \downarrow & & \downarrow \\
& & \mathcal{V}^\otimes & \xrightarrow{f} & \mathcal{W}^\otimes
\end{array}$$

, which is also true, since pullback in $\text{coC}/_{\text{Ass}}$ can be computed in $\text{Cat}/_{\text{Ass}}$, so there is a unique induced morphism in $\text{coC}/_{\text{Ass}} \mathcal{N}^{\otimes} \rightarrow \mathcal{M}^{\otimes}$ making the diagram commute, thus preserve coCartesian edges, and is a linear functor.

- Thus $f_!$ is locally Cartesian in $\text{LMod} \rightarrow \text{Mon}$, and thus Cartesian.

□

5.3 Symmetric monoidal structure

By [15, lemma 7.10], the coCartesian fibration $\text{PreCat} \rightarrow \text{Op}^{ns}$ preserves finite products; thus, there is an operad map:

$$\text{PreCat}^{\times} \rightarrow (\text{Op}^{ns})^{\times}$$

Moreover, the operad map is a coCartesian fibration by [15, remark 7.11].

Remark 5.3.1. Let $\boxtimes$ denote the product in PreCat , which, by [15, lemma 7.10], we can describe $F \boxtimes G$ for categorical algebras $F : \text{Ass}_X \rightarrow \mathcal{V}^{\otimes}$ and $G : \text{Ass}_Y \rightarrow \mathcal{W}^{\otimes}$ as:

$$\text{Ass}_{X \times Y} \cong \text{Ass}_X \times_{\text{Ass}} \text{Ass}_Y \rightarrow \mathcal{V}^{\otimes} \times_{\text{Ass}} \mathcal{W}^{\otimes}$$

, which is a categorical algebra in $\mathcal{V} \times \mathcal{W}^{\otimes}$.

Likewise, $\text{Enr} \rightarrow \text{Op}^{ns}$ also preserves finite products; thus, it can be enhanced into an operad map:

$$\text{Enr}^{\times} \rightarrow (\text{Op}^{ns})^{\times}$$

.

Given a symmetric monoidal category $\mathcal{V}$, it can be regarded as a commutative algebra object in Op^{ns} , specified by a map:

$$\{\mathcal{V}^{\otimes}\} : \text{Comm}^{\otimes} \rightarrow (\text{Op}^{ns})^{\times}$$

Notation 5.3.2. Given a symmetric monoidal category $\mathcal{V}$,

- (1) Let $\text{Alg}_{\text{cat}}(\mathcal{V})^{\otimes}$ denote the fiber product in Op :

$$\begin{array}{ccc} \text{Alg}_{\text{cat}}(\mathcal{V})^{\otimes} & \longrightarrow & \text{PreCat}^{\times} \\ \downarrow \lrcorner & & \downarrow \\ \text{Comm}^{\otimes} & \xrightarrow{\{\mathcal{V}^{\otimes}\}} & (\text{Op}^{ns})^{\times} \end{array}$$

- (2) Let $\text{Enr}_{\mathcal{V}}^{\otimes}$ denote the fiber product in Op :

$$\begin{array}{ccc} (\text{Cat}^{\mathcal{V}})^{\otimes} & \longrightarrow & \text{Enr}^{\times} \\ \downarrow \lrcorner & & \downarrow \\ \text{Comm}^{\otimes} & \xrightarrow{\{\mathcal{V}^{\otimes}\}} & (\text{Op}^{ns})^{\times} \end{array}$$

- (3) By [17, proposition 4.26], $\text{PLMod}^\times \rightarrow \text{Mon}^\times$ is also a coCartesian fibration, thus let $\text{PLMod}_\mathcal{V}^\otimes$ denote the fiber product in Op :

$$\begin{array}{ccc} \text{PLMod}_\mathcal{V}^\otimes & \longrightarrow & \text{PLMod}^\times \\ \downarrow & \lrcorner & \downarrow \\ \text{Comm}^\otimes & \xrightarrow{\{\mathcal{V}^\otimes\}} & \text{Mon}^\times \end{array}$$

Proposition 5.3.3. *Given a symmetric monoidal category $\mathcal{V}$, there is a symmetric monoidal adjunction:*

$$\text{loc}_\mathcal{V} : \text{Alg}_{\text{cat}}(\mathcal{V})^\otimes \rightleftarrows (\text{Cat}^\mathcal{V})^\otimes : i_\mathcal{V}$$

Moreover, given a symmetric monoidal adjunction

$$L : \mathcal{V}^\otimes \rightleftarrows \mathcal{W}^\otimes : R$$

Transfer of enrichment along L is again symmetric monoidal; thus, there is a symmetric monoidal adjunction:

$$L_*^{\text{cat}} : \text{Alg}_{\text{cat}}(\mathcal{V})^\otimes \rightleftarrows \text{Alg}_{\text{cat}}(\mathcal{W})^\otimes : R_*^{\text{cat}}; L_* : \text{Cat}^\mathcal{V} \rightleftarrows \text{Cat}^\mathcal{W} : R_*$$

. And there is a commutative diagram:

$$\begin{array}{ccc} \text{Alg}_{\text{cat}}(\mathcal{V})^\otimes & \xrightarrow{L_*^{\text{cat}}} & \text{Alg}_{\text{cat}}(\mathcal{W})^\otimes \\ \text{loc}_\mathcal{V} \downarrow & & \downarrow \text{loc}_\mathcal{W} \\ (\text{Cat}^\mathcal{V})^\otimes & \xrightarrow{L_*} & (\text{Cat}^\mathcal{W})^\otimes \end{array}$$

Proof. Consider the localization

$$\text{loc} : \text{PreCat}^\times \times_{(\text{Op}^{ns})^\times} \text{Mon}^\times \rightleftarrows \text{Enr}^\times \times_{(\text{Op}^{ns})^\times} \text{Mon}^\times : i$$

. First, we show that loc preserves all coCartesian liftings over $\text{Mon}^\times$: it suffices to show that it preserves the coCartesian liftings over the active map in $\text{Mon}^\times$: consider an active map $(\mathcal{V}, \mathcal{W}) \rightarrow \mathcal{V} \times \mathcal{W}^\otimes$, given localization maps of categorical algebra $\mathcal{M}$ in $\mathcal{V}$ and $\mathcal{N}$ in $\mathcal{W}$, it remains to show that the induced map $\mathcal{M} \boxtimes \mathcal{N} \rightarrow (L\mathcal{M} \boxtimes L\mathcal{N})$ is a local equivalence (fully faithful and essential surjective) in PreCat , which is true by the description of $\boxtimes$ in [remark 5.3.1](#).

Thus the induced map by taking pullback of $\text{PreCat}^\times \times_{(\text{Op}^{ns})^\times} \text{Mon}^\times$ along $\{\mathcal{V}^\otimes\} : \text{Comm}^\otimes \rightarrow \text{Mon}^\times$ is symmetric monoidal, and the adjunction between underlying categories can be enhanced into a symmetric monoidal adjunction:

$$\text{loc}_\mathcal{V} : \text{Alg}_{\text{cat}}(\mathcal{V}) \rightleftarrows (\text{Cat}^\mathcal{V})^\otimes$$

Let $[1] \times \text{Comm}^\otimes \rightarrow (\text{Op}^{ns})^\times$ encode the symmetric monoidal functor L , then by pullback

along the coCartesian fibrations $loc : \text{PreCat}^\times$ and $\text{Enr}^\times$ over $(\text{Op}^{ns})^\times$, the acquired coCartesian fibration over $[1] \times \text{Comm}^\otimes$ classifies the symmetric monoidal functor L_*^{cat} and L_* . And the right adjoints R_*^{cat} and R_* can be enhanced into lax symmetric monoidal functors by [25, corollary 7.3.2.7].

□

Notation 5.3.4. (1) Let $\text{PPreCat}^\otimes$ denote the full suboperad of the fiber product $\widehat{\text{PreCat}}^\times \times_{\widehat{\text{Mon}}^\times} \text{Mon}^\times$ along $\mathcal{P}(-) : \text{Mon}^\times \rightarrow \widehat{\text{Mon}}^\times$, consisting of those categorical algebras with small underlying spaces of objects.

(2) Let $\text{PEnr}^\otimes$ denote the full suboperad of the fiber product $\widehat{\text{Enr}}^\times \times_{(\widehat{\text{Op}^{ns}})^\times} \text{Mon}^\times$ along $\mathcal{P}(-) : \text{Mon}^\times \rightarrow \widehat{\text{Mon}}^\times$, consisting of those enriched categories with small underlying spaces of objects.

By [15, corollary 6.13] and [15, lemma 7.14], there is:

Proposition 5.3.5. $\text{PPreCat}^\otimes \rightarrow \text{Comm}^\otimes$ is a Cartesian symmetric monoidal category, and moreover, there is a symmetric monoidal embedding $\text{PLMod}^\times \hookrightarrow \text{PPreCat}^\times$ over $\text{Mon}^\times$ which essential image consists of complete categorical algebras. Thus the symmetric monoidal full subcategory $\text{PEnr}^\otimes \hookrightarrow \text{PPreCat}^\otimes$ is also of Cartesian symmetric monoidal structure, and there is an induced symmetric monoidal equivalence $\text{PLMod}^\times \cong \text{PEnr}^\otimes$.

Now consider the lax symmetric monoidal inclusion $\text{LMod}^\times \rightarrow \text{PLMod}^\times$ over $\text{Mon}^\times$; we have the following corollary by [15, theorem 7.21]:

Corollary 5.3.6. Given a symmetric monoidal category $\mathcal{V}$, there is a lax symmetric monoidal inclusion $\text{LMod}_{\mathcal{V}}(\text{Cat})^\otimes \rightarrow \text{PLMod}_{\mathcal{V}}^\otimes \cong (\text{Cat}^{\mathcal{P}(\mathcal{V})})^\otimes$. Moreover, given a symmetric monoidal functor $f : \mathcal{V} \rightarrow \mathcal{W}$, the above inclusion is contravariant:

$$\begin{array}{ccc} \text{LMod}_{\mathcal{V}}(\text{Cat}) & \longrightarrow & \text{Cat}^{\mathcal{P}(\mathcal{V})} \\ \uparrow & & \uparrow \\ \text{LMod}_{\mathcal{W}}(\text{Cat}) & \longrightarrow & \text{Cat}^{\mathcal{P}(\mathcal{W})} \end{array}$$

where the left vertical map is the forgetful functor along $\mathcal{V} \rightarrow \mathcal{W}$ and the right vertical map is the right adjoint of the transfer of enrichment along $\mathcal{P}(f) : \mathcal{P}(\mathcal{V}) \rightarrow \mathcal{P}(\mathcal{W})$.

Proof. • Consider the symmetric monoidal inclusion $\text{LMod}^\times \rightarrow \text{PLMod}^\times \rightarrow \text{PPreCat}$ over $\text{Mon}^\times$. By taking the pullback along $\{\mathcal{V}^\otimes\} : \text{Comm}^\otimes \rightarrow \text{Mon}^\times$, there is a lax symmetric monoidal inclusion

$$\text{LMod}_{\mathcal{V}}(\text{Cat})^\otimes \rightarrow \text{PLMod}_{\mathcal{V}}^\otimes \cong (\text{Cat}^{\mathcal{P}(\mathcal{V})})^\otimes \hookrightarrow \text{PreCat}_{\mathcal{V}}^\otimes$$

- Since $\text{PLMod} \cong \text{PEnr}$ is Cartesian over Mon by proposition 5.2.4, and also coCartesian since it's the pullback of the coCartesian fibration $\text{Enr} \rightarrow \text{Op}^{ns}$, thus $\text{PLMod} \rightarrow \text{Mon}$ is a biCartesian fibration.

- Let $\eta : [1] \times \text{Comm}^\otimes \rightarrow \text{Mon}^\times$ encode the symmetric monoidal functor $f : \mathcal{V} \rightarrow \mathcal{W}$, then consider the pullback along the morphism $\eta : [1] \times \text{Comm}^\otimes \rightarrow \text{Mon}^\times$. This gives rise to a morphism F between bifibrations over $[1] \times \text{Comm}^\otimes$ (Cartesian over $[1]$ and coCartesian over $\text{Comm}^\otimes$), as illustrated in the following diagram:

$$\begin{array}{ccccc}
 M & \xrightarrow{F} & N & & \\
 \downarrow p \wr & \searrow & \downarrow q \wr & \searrow & \\
 & & \text{LMod}^\times & \xrightarrow{i} & \text{PLMod}^\times \\
 & & \downarrow & & \downarrow \\
 [1] \times \text{Comm}^\otimes & \xrightarrow{\quad} & [1] \times \text{Comm}^\otimes & \xrightarrow{\quad} & \text{Mon}^\times \\
 & \searrow & \downarrow & \searrow & \\
 & & \text{Mon}^\times & \xrightarrow{\quad} & \text{Mon}^\times
 \end{array}$$

- The restriction of F to the fibers over $\{0\} \times \text{Comm}^\otimes \hookrightarrow [1] \times \text{Comm}^\otimes$ is the operad map $\text{LMod}_\mathcal{V}^\otimes \rightarrow \text{PLMod}_\mathcal{V}^\otimes$.
- The restriction of F to the fibers over $\{1\} \times \text{Comm}^\otimes \hookrightarrow [1] \times \text{Comm}^\otimes$ is the lax symmetric monoidal functor $R : \text{LMod}_\mathcal{W}(\text{Cat})^\otimes \hookrightarrow \text{PLMod}_\mathcal{W}^\otimes$.
- M and N are **biCartesian fibrations** over $[1]$ and **coCartesian fibrations** over $\text{Comm}^\otimes$ since pullback preserves Cartesian and coCartesian fibrations.
- $F : M \rightarrow N$ is a functor preserving Cartesian edges over $[1]$ and preserves the coCartesian edges over the inert maps in $\text{Comm}^\otimes$.
- Consider the fibration $M \rightarrow [1] \times \text{Comm}^\otimes$ which is biCartesian fibration over $[1]$ and coCartesian fibration over $\text{Comm}^\otimes$, by projection to $[1]$, we get an adjunction relative to $\text{Comm}^\otimes$ by [14, 2.2.4] :

$$f_! : \text{LMod}_\mathcal{V}^\otimes \rightleftarrows \text{LMod}_\mathcal{W}(\text{Cat})^\otimes : f^!$$

, where $f_!$ is the symmetric monoidal ∞ -functor $(-) \otimes_\mathcal{V} \mathcal{W}$, and $f^!$ is the forgetful functor along $\mathcal{V} \rightarrow \mathcal{W}$.

- Consider the fibration $N \rightarrow [1] \times \text{Comm}^\otimes$, which is a biCartesian fibration over $[1]$ and a coCartesian fibration over $\text{Comm}^\otimes$. By projection to $[1]$, we get an adjunction relative to $\text{Comm}^\otimes$ by [14, 2.2.4] :

$$f_* : \text{PLMod}_\mathcal{V}^\otimes \rightleftarrows \text{PLMod}_\mathcal{W}^\otimes : f^*$$

, where f_* is the transfer of enrichment along $\mathcal{P}(f)$ under the equivalence $\text{PLMod}_\mathcal{V}^\otimes \cong (\text{Cat}^{\mathcal{P}(\mathcal{V})})^\otimes$ and $(\text{PLMod}_\mathcal{W})^\otimes$.

- Since F preserves the Cartesian edges over $[1]$, by [14, 2.2.4], the straightening gives us a morphism $\eta : [1] \rightarrow \text{Fun}([1]^{op}, \text{Cat}_{/\text{Comm}^\otimes})$, this is equivalent to a functor $[1] \times [1]^{op} \rightarrow \text{Cat}_{/\text{Comm}^\otimes}$, as illustrated in the following diagram in $\text{Cat}_{/\text{Comm}^\otimes}$:

$$\begin{array}{ccc}
\mathrm{LMod}_{\mathcal{W}}^{\otimes} & \longrightarrow & \mathrm{PLMod}_{\mathcal{W}}^{\otimes} \\
\downarrow f^! & & \downarrow f^* \\
\mathrm{LMod}_{\mathcal{V}}^{\otimes} & \longrightarrow & \mathrm{PLMod}_{\mathcal{V}}^{\otimes}
\end{array}$$

□

5.4 Loop and deloop

Consider the functor $\mathrm{PreCat} \rightarrow \mathrm{Op}^{ns} \times \mathrm{Spc}$ which is coCartesian fibration over Op^{ns} and Cartesian over Spc , by taking the fiber over $* \in \mathrm{Spc}$, by the construction of Quiv in [15, notation 4.37], $\mathrm{PreCat} \times_{\mathrm{Spc}} *$ is the coCartesian fibration over Op^{ns} classifying the functor:

$$\mathrm{Alg}_{\mathrm{Ass}}(-) : \mathrm{Op}^{ns} \rightarrow \mathrm{Cat}$$

which is the right adjoint of $(-) \times \mathrm{Ass} : \mathrm{Cat} \rightarrow \mathrm{Op}^{ns}$.

Notation 5.4.1. Let Alg denote the above fiber product $\mathrm{PreCat} \times_{\mathrm{Spc}} *$.

Moreover, one can consider the symmetric monoidal structure on the above construction: Given that the fiber embedding $\mathrm{PreCat}_0 \hookrightarrow \mathrm{PreCat}$ preserves products in PreCat by the description of $\boxtimes$ in remark 5.3.1, there is a symmetric monoidal functor

$$\mathrm{Alg}^{\times} \hookrightarrow \mathrm{PreCat}^{\times}$$

over $(\mathrm{Op}^{ns})^{\times}$.

Construction 5.4.2. Given a symmetric monoidal category $\mathcal{V}$, by pullback along $\{\mathcal{V}^{\otimes}\} : \mathrm{Comm}^{\otimes} \rightarrow (\mathrm{Op}^{ns})^{\times}$, there is a symmetric monoidal functor:

$$B : \mathrm{Alg}(\mathcal{V})^{\otimes} \hookrightarrow \mathrm{PreCat}_{\mathcal{V}}^{\otimes} \xrightarrow{\mathrm{loc}_{\mathcal{V}}} \mathrm{Cat}^{\mathcal{V}}$$

where $\mathrm{Alg}(\mathcal{V})^{\otimes}$ is the pullback of $\mathrm{Alg}^{\times} \rightarrow (\mathrm{Op}^{ns})^{\times}$ along $\{\mathcal{V}\}$.

Remark 5.4.3. By [30, theorem 0.1], the symmetric monoidal category $\mathrm{Alg}(\mathcal{V})^{\otimes}$ in the above construction is equivalent to the commutative algebra object structure on the image of $\mathcal{V} \in \mathrm{CAlg}(\mathrm{Op}^{ns})$ in Cat under the lax symmetric monoidal functor $\mathrm{Alg}_{\mathrm{Ass}}(-) : \mathrm{Op}^{ns} \rightarrow \mathrm{Cat}$.

Theorem 5.4.4. [10, theorem 6.32] *Given a symmetric monoidal category $\mathcal{V}$, the symmetric monoidal functor B in construction 5.4.2 is fully faithful, and admits a right adjoint Ω .*

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